Portfolio Substitution and the Effects of Bank Capital Regulation

Laura Nicolae
Harvard

November 2026
(most recent version)

Abstract

A large literature finds that bank capital requirements reduce lending, but these studies generally estimate lending responses to regulations that change required equity for multiple bank assets simultaneously. For example, a 1-percentage-point increase in a bank's risk-based capital requirement raises the required equity on risk-weight-1 loans by 1 percentage point and on risk-weight-0.2 securities by 0.2 percentage points. Evaluating counterfactual policies that impose a new pattern of equity requirements across assets, such as changing a single asset's risk weight, requires estimating the response of an asset's supply to its own regulatory cost while holding other assets' costs fixed. I develop a portfolio-allocation model and empirical method to estimate each asset's own-cost elasticity of supply using changes in bank-level capital requirements. The model uses the fact that portfolio shares sum to one to express each asset's own-cost response and portfolio substitution in terms of the same elasticities. I recover positive supply elasticities for every asset category, with securities holdings responding more to regulatory costs than loans do. Applying these elasticities to post-crisis requirements, I ask whether stronger capital regulation explains large banks' shift from lending to securities since the financial crisis. Capital regulation accounts for about 28% of the aggregate shift toward securities and 35% of the cross-sectional convergence in large banks' average asset risk weights since 2011.

1 Introduction

Bank capital requirements are regulators' principal tool for protecting financial stability. Capital requirements require banks to finance a minimum share of their assets with equity, reducing banks' incentives to take risks. But stronger capital requirements also reduce bank lending by raising banks' funding costs (e.g., Aiyar et al., 2014; Bridges et al., 2014; Gropp et al., 2019; Berrospide and Edge, 2019; Juelsrud and Wold, 2020; Favara et al., 2021; Blattner et al., 2023).

Banks' funding costs can rise when banks finance a larger share of their assets with equity, in contrast to the Modigliani–Miller benchmark in which a firm's financing mix leaves its total funding cost unchanged, because deposits and other safe bank debt provide liquidity services that investors value, allowing banks to borrow at lower rates (Gorton and Pennacchi, 1990; Stein, 2012; DeAngelo and Stulz, 2015; Gorton and Winton, 2017). Understanding how capital requirements affect banks' portfolio composition is therefore central to assessing the costs and benefits of capital regulation.

Since the financial crisis, regulators have strengthened capital requirements for large U.S. banks, while these banks have simultaneously shifted their portfolios away from lending and toward securities (Hanson et al., 2024). A prominent explanation for banks' declining role in credit provision is that stronger bank regulation has discouraged bank lending and encouraged lending by nonbanks (Buchak et al., 2018; Irani et al., 2021; Lee et al., 2024).

Because past changes in bank capital requirements have typically altered the equity requirements on multiple assets at once, most studies estimate how an asset's supply responds when its own equity requirement and other assets' requirements change together. For example, a 10% risk-based capital requirement at the bank level requires no equity financing for risk-weight-0 securities and requires 10% equity financing for risk-weight-1 loans. The supplementary leverage ratio (SLR) requires 5% equity financing for both assets, imposing a higher equity requirement on securities and a lower requirement on loans than the risk-based rule. Research shows that banks subject to the SLR reduce their holdings and intermediation of securities (Duffie and Krishnamurthy, 2016; Favara et al., 2022; Bräuning and Stein, 2024). Since the SLR changes the regulatory cost of securities and loans simultaneously, the estimated response reflects both channels; separating them requires variation in the cost of one asset class holding the other fixed, which is the focus of this paper.

More generally, predicting banks' response to a counterfactual policy that imposes a different pattern of regulatory penalties across assets from those observed under current policies requires separating how an asset responds to its own regulatory cost from how it responds to changes in other assets' costs. To solve this problem, I develop a portfolio-

allocation model and empirical method to estimate each asset's own-cost elasticity of supply using past changes in bank-level capital requirements. In the model, each asset's own-cost elasticity governs both its direct response to its own regulatory cost and its substitution response to changes in other assets' regulatory costs. The portfolio adding-up identity links these responses across assets, so each asset's substitution response also depends on the other assets' elasticities.

My method recovers positive supply elasticities for every asset category and finds that securities respond more to regulatory costs than loans do. This is intuitive because banks can purchase securities in liquid markets, while expanding lending requires finding and screening borrowers. The model's predictions match out-of-sample bank responses to changes in capital requirements.

The resulting supply elasticity estimates are structural parameters useful for counterfactual policy analysis. I use them to ask whether stronger capital regulation explains large banks' shift from lending to securities since the financial crisis. Applying the estimated asset supply elasticities to post-crisis capital requirements, I find that capital regulation accounts for about 28% of the aggregate shift from lending toward securities and 35% of the cross-sectional convergence in large banks' average asset risk weights since 2011. Stronger capital requirements thus explain part, but not most, of the decline in large banks' lending shares and the convergence in their average asset risk weights.

This paper begins by developing a model of bank portfolio allocation across multiple types of loans and securities. For each asset, the model imposes the same local curvature in net-return functions across banks while allowing net-return levels to differ, so banks share a common local own-cost elasticity of supply for a given asset but can earn different returns on the same asset. In this model, holding return schedules and bank sizes fixed, an asset loses portfolio share when its regulatory cost rises relative to the average regulatory cost of all the bank's assets, weighted by their supply elasticities. The intuition follows from the portfolio adding-up identity: since portfolio shares sum to one, an expansion in one asset's portfolio share requires a contraction in other assets' shares. The same own-cost elasticity governs both an asset's direct response to its regulatory cost and its substitution response to changes in other assets' regulatory costs.

A feature of the model is that it is straightforward to estimate in the data. I estimate asset supply elasticities using changes in bank-level capital requirements that alter assets' regulatory penalties by different amounts. The sample covers 2011–2025 and includes 28 domestic bank holding companies ("banks") with more than $100 billion in assets in at least one quarter during this period. The analysis covers banking-book assets only, excluding the trading book. I match quarterly bank balance-sheet data from Form FR Y-9C to the

standardized asset risk weights prescribed by U.S. capital regulation (12 CFR Part 217, Subpart D), grouping assets into four categories: 0, 0.2, 0.5, and 1. I combine these risk weights with bank-level capital requirements to measure the share of an asset that must be funded by equity for each bank–asset category pair in each quarter. I estimate each asset's supply elasticity using time-series variation in the equity required per dollar for a bank–asset pair across multiple regulatory changes.

My method identifies asset supply elasticities using two sources of regulatory variation: increases in risk-based capital requirements and switches in banks' binding capital rule. The risk-based capital (RBC) rule requires more equity against assets with higher regulatory risk weights, while the supplementary leverage ratio (SLR) requires the same equity share for all assets, regardless of asset risk. Large banks must satisfy both rules; a bank's binding rule is the rule that requires the most equity funding for its portfolio. When the RBC binds, an increase in the bank's required RBC ratio raises its required equity more for assets with higher risk weights. For example, a 1-percentage-point increase in the bank's required RBC ratio raises the equity requirement on risk-weight-1 unsecured loans by 1 percentage point, on risk-weight-0.5 mortgages by 0.5 percentage points, and so on. Since 2013, the Fed has raised RBC requirements by different amounts for different banks over time. Switches between binding rules can instead raise a bank's required equity on some assets while lowering it on others. For example, the introduction of the SLR caused custodian and dealer banks to switch from being RBC-bound to SLR-bound, raising their effective equity requirement on securities and lowering it on loans; subsequent increases in RBC requirements caused several of these banks to switch back.

I use both sources of variation to compare 4–5-year portfolio-share growth across assets facing different changes in required equity within the same bank, relative to portfolio-share growth in the same assets at other banks. I estimate asset supply elasticities jointly by nonlinear least squares, controlling for asset–date–horizon fixed effects to absorb shocks to assets' net returns that are common to all banks holding the same asset. Identification relies on a parallel-trends assumption, as in difference-in-differences. The assumption is that, absent changes in regulation, differences in the portfolio-share growth across assets at the same bank would have been the same as those at other banks.

After correcting for portfolio substitution, I find that increasing an asset's equity requirement reduces its portfolio share for every asset category, holding other assets' equity requirements fixed. These substitution-corrected portfolio-share responses are more negative than the uncorrected estimates for every asset, with the response of the risk-weight-0.5 asset (mostly first-lien mortgages) changing from positive to negative under the substitution correction. Without controlling for portfolio substitution, a regression of portfolio shares

on equity requirements would suggest that mortgages gain portfolio share when their equity requirement rises.

I also find that securities respond more to their own regulatory costs than loans do, consistent with their greater liquidity and homogeneity. This is intuitive because a bank can purchase securities in a standardized, liquid market, whereas expanding lending requires finding and screening additional borrowers and may also require building borrower relationships and developing expertise in unfamiliar markets. These factors limit how easily a bank can increase lending when its regulatory cost falls.

I evaluate the estimated asset supply elasticities out of sample using the 2020 and 2021 Stress Capital Buffer changes, which were excluded from the estimation. The model correctly predicts the direction of banks' portfolio-share responses in three of the four asset categories.

To illustrate how the estimated supply elasticities translate into changes in portfolio shares, consider imposing a 5% SLR equity requirement on a bank previously bound by a 10% RBC equity requirement, so that the SLR becomes binding. The SLR imposes a flat equity requirement across all assets regardless of risk, so it raises required equity on low-risk-weight securities and lowers it on high-risk-weight loans. Applying the estimated supply elasticities to these asset-specific changes in required equity implies a decline in risk-weight-0 assets (mostly Treasury securities and some reverse repo) of about 3.2 percentage points and an increase in banks' unsecured-loan holdings (e.g., credit card loans and C&I) of about 1.7 percentage points.

Beyond these portfolio responses, I use the estimated supply elasticities to infer differences in banks' net returns that account for their persistent specialization after controlling for equity requirements. I use data on banks' portfolio shares in each asset to recover bank-specific net-return levels on each asset, using the estimated supply elasticities to convert cross-bank differences in portfolio shares into implied differences in net returns. The estimates reveal comparative advantages in lending for traditional banks and in securities and reverse repos for dealers and custodians, each worth several percentage points in annual net returns.

Finally, I examine the contribution of stronger capital requirements to large banks' shift from lending to securities since 2011, an issue central to the debate over banks' declining role in credit provision (Hanson et al., 2024; Buchak et al., 2018; Irani et al., 2021; Lee et al., 2024; Buchak et al., 2024). Using the actual equity requirements that large banks faced on every asset risk-weight category in each quarter since 2011, I compute banks' model-implied portfolio shares using the estimated asset supply elasticities. This partial-equilibrium exercise holds banks' sizes and assets' net-return schedules fixed. I find that stronger capital requirements account for about 28% of large U.S. banks' observed shift from lending to

securities between 2011 and 2024. I also estimate the contribution of post-crisis capital requirements to the convergence in large banks' average asset risk weights since 2011. Since each bank tends to be bound by the rule that most strongly penalizes its core assets, risk-based capital requirements encourage lending specialists to expand securities holdings, while the leverage requirement encourages dealers and custodians to expand lending (Greenwood et al., 2017). I find that post-crisis capital requirements account for about 35% of the observed convergence in banks' average portfolio risk weights.

The remainder of the paper proceeds as follows. Section 2 develops the model of banks' choice of portfolio allocation and equity ratio subject to regulatory capital requirements and derives the estimating equation. Section 3 describes the data, regulatory variation, and empirical implementation. Section 4 presents the own-cost supply elasticity estimates, examines alternative specifications, validates the model's predictions using out-of-sample changes in capital requirements, and recovers banks' comparative advantages. Section 5 estimates the contribution of post-crisis capital requirements to large banks' shift from lending toward securities and to the convergence in their portfolio compositions. Section 6 concludes.

Contribution to the literature. Most studies of bank capital requirements estimate how lending responds when equity requirements on loans and other assets change together. My paper estimates how an asset's supply responds to its own regulatory cost when other assets' regulatory costs are held fixed.

The model's portfolio-substitution mechanism also helps interpret existing evidence that balance-sheet constraints limit banks' market intermediation. This literature links balance-sheet constraints such as leverage regulation to reduced repo and Treasury-market intermediation and persistent deviations from covered interest parity. For example, Duffie and Krishnamurthy (2016) explain how the SLR discourages repo intermediation and impairs monetary-policy pass-through, while He et al. (2022) model how regulatory balance-sheet constraints limit Treasury purchases and repo financing. Consistent with these mechanisms, Duffie et al. (2023) find that Treasury-market liquidity deteriorates when dealer balance-sheet utilization is high, and Bräuning and Stein (2024) find that relaxing dealer constraints increases Treasury-market intermediation and improves liquidity. Du et al. (2018) link persistent covered-interest-parity deviations to regulatory constraints. My estimates show that the SLR shifts banks' banking-book portfolios away from securities and toward lending. The model illustrates how substitution toward lending can reinforce the direct effect of higher equity requirements on securities holdings.

My paper also contributes to the literature on banks' portfolio allocation and substitution across assets. Furfine (2001) estimates a dynamic model of bank portfolio choice under capital requirements and regulatory monitoring. Wang et al. (2022) structurally estimate

banks' joint lending, securities, and funding choices under capital and reserve requirements to study monetary policy transmission. Li et al. (2026) model banks' joint choices of securities holdings, corporate lending, lending to nonbanks, and deposit funding, subject to a balance-sheet constraint. My estimation framework is most closely related to Diamond et al. (2024), who use instruments for loan demand, deposit demand, and reserve supply to estimate how banks' marginal costs depend on their balance-sheet composition. My approach instead uses variation in regulatory costs and the portfolio adding-up identity to separately estimate own-cost supply elasticities and substitution across securities and loan categories.

My framework studies how differences in asset risk weights shape allocation across a bank's portfolio as a whole, complementing studies of how loan-specific capital requirements shape allocation within corporate or mortgage lending (Behn et al., 2016; Mayordomo and Rodríguez-Moreno, 2018; Dietsch et al., 2020; Plosser and Santos, 2024). Within corporate lending, Fraisse et al. (2020) use differences in capital requirements across bank–firm pairs to show that higher loan-specific requirements reduce lending. Within mortgage lending, Benetton et al. (2021) use variation in capital requirements across loans in the same lender's mortgage book to show that lower requirements reduce mortgage rates; Benetton (2021) estimates the effects of capital requirements on mortgage origination costs and lender competition.

This paper also contributes to the debate over banks' declining role in credit provision. Begenau and Landvoigt (2022) show that higher bank capital requirements expand shadow banking, while Acharya et al. (2026) argue that bank regulation encourages banks to replace direct lending to borrowers with financing for nonbanks that lend to those borrowers. Sundaresan and Xiao (2024) identify a related effect of liquidity regulation on banks' portfolios: liquidity regulations encourage banks to hold cash and liquid securities at the expense of lending. Chernenko et al. (2025) challenge the capital-requirements explanation for private credit's growth and emphasize bank supervision and the regulatory treatment of indirect lending. Technology and funding offer additional explanations: Buchak et al. (2024) find that improvements in securitization made securities-based financing more attractive to borrowers relative to bank balance-sheet loans, while savers' growing preference for securities over deposits reduced bank funding. Hanson et al. (2024) emphasize the role of deposit growth in banks' declining loan shares. Koont and Walz (2026) find that banks' lending response to deposit inflows has weakened as balance-sheet expansion has become more costly. My exercise isolates the contribution of capital requirements to portfolio composition, holding bank size fixed, and finds that stronger capital requirements explain about 28% of large banks' shift from lending toward securities.

Finally, my estimation builds on demand systems that link own-price and cross-price re-

sponses through budget constraints and structural assumptions (Brainard and Tobin, 1968; Deaton and Muellbauer, 1980). Koijen and Yogo (2019) develop an instrumental-variables approach to estimating asset demand. Fuchs et al. (2023) show that misspecified substitution can distort estimated own-price elasticities. Haddad et al. (2026) show that, under conditional homogeneity of substitution, cross-sectional designs identify relative elasticities but leave substitution patterns needed for broader counterfactuals unidentified. I apply this approach to banks' asset supply, using regulatory changes that move several assets' funding costs simultaneously. I assume that each asset's net-return function depends only on its own portfolio share. Together with the portfolio adding-up identity, this assumption links an asset's direct response to required equity to its substitution responses to other assets' costs, allowing me to estimate supply elasticities jointly from portfolio-share responses.

2 Model and Estimation

The model links banks' portfolio choices under capital requirements to an estimating equation for asset-level own-cost supply elasticities. I first describe the bank's problem and optimality conditions, then use the portfolio adding-up identity to link own-cost responses to portfolio substitution and derive the estimating equation.

2.1 Bank Portfolio Choice

Banks choose their portfolio composition and equity ratios subject to risk-based capital (RBC) requirements and a supplementary leverage ratio (SLR). Under RBC, assets with higher legally prescribed risk weights w_a require more equity financing, and banks with higher required ratios of equity to risk-weighted assets ℓ_{bt}^{RBC} must finance more of each asset with equity. The SLR requires the same equity financing share for every asset, with ℓ_{bt}^{SLR} denoting the minimum ratio of equity to total assets.[1]

Bank b chooses portfolio shares s_{abt} in assets a and an equity-to-assets ratio E_{bt} in period t, holding total assets fixed. The index a covers all assets in the bank's portfolio.[2] The function $r_{abt}(s_{abt})$ gives the bank's expected income from asset a, net of operating and debt funding costs, divided by the bank's total assets. Net returns are separable across assets. I assume

[1]The statutory SLR measures Tier 1 capital as a share of total leverage exposure, which includes off-balance-sheet exposures. Here its required ratio is converted to a requirement per dollar of assets, holding leverage exposure relative to assets fixed.

[2]In the empirical implementation, assets without an observed standardized risk weight are grouped into an outside good and retained in the portfolio-share denominator. They are included in the asset index here. The baseline estimation holds their combined portfolio share fixed, so changes in the shares of the remaining asset categories sum to zero; Appendix C relaxes this restriction.

that r_{abt} is continuously differentiable and strictly concave in the asset's portfolio share. Diminishing marginal net returns can reflect declining marginal revenues, rising marginal operating costs, or both. For example, expanding lending may require paying higher wages to recruit additional loan officers or offering lower loan rates to attract additional borrowers.

Banks' funding costs can rise when banks finance a larger share of their assets with equity, in contrast to the Modigliani–Miller benchmark in which a firm's financing mix leaves its total funding cost unchanged, because banks' deposits and other safe, short-term debt provide liquidity services that investors value, allowing banks to issue these claims at lower financing costs (Gorton and Pennacchi, 1990; Stein, 2012; DeAngelo and Stulz, 2015; Gorton and Winton, 2017). The parameter c_E measures the additional cost per dollar of financing assets with equity rather than debt.

The bank maximizes expected profit subject to portfolio and capital constraints:

$$\begin{aligned} &\max_{\{s_{abt}\},E_{bt}} \quad \sum_a r_{abt}(s_{abt}) - c_E E_{bt} \\ &\text{subject to} \quad \sum_a s_{abt} = 1, \quad s_{abt} \geq 0, \quad E_{bt} \geq \ell_{bt}^{RBC} \sum_a w_a s_{abt}, \quad E_{bt} \geq \ell_{bt}^{SLR} \sum_a s_{abt}. \end{aligned} \tag{1}$$

I derive the portfolio first-order conditions for an interior allocation, with $s_{abt} > 0$ for every asset. The Lagrangian incorporates the portfolio adding-up identity with multiplier η_{bt} and the capital constraints with multipliers $\lambda_{bt}^{RBC}, \lambda_{bt}^{SLR} \geq 0$:

$$\begin{aligned} \mathcal{L}_{bt} = \sum_a r_{abt}(s_{abt}) - c_E E_{bt} + \eta_{bt} \left(1 - \sum_a s_{abt}\right) \\ + \lambda_{bt}^{RBC} \left(E_{bt} - \ell_{bt}^{RBC} \sum_a w_a s_{abt}\right) + \lambda_{bt}^{SLR} \left(E_{bt} - \ell_{bt}^{SLR} \sum_a s_{abt}\right). \end{aligned}$$

The first-order condition for equity is

$$\lambda_{bt}^{RBC} + \lambda_{bt}^{SLR} = c_E.$$

Because equity is more costly than debt, the bank chooses the minimum equity ratio that satisfies both capital requirements. Ruling out a tie, the rule with the higher required equity ratio binds and the other rule is slack. By complementary slackness, the slack requirement's multiplier is zero, so the binding requirement's multiplier equals c_E.

The marginal equity requirement can differ across banks for the same asset because different capital rules bind. Suppose Capital One holds only credit-card loans and Goldman Sachs

holds only Treasuries.[3] Both banks face a 10% RBC requirement and a 5% SLR requirement. A portfolio consisting entirely of credit-card loans, with a risk weight of 1, requires equity equal to 10% of assets under RBC and 5% under SLR. A portfolio consisting entirely of Treasuries requires zero equity under RBC but 5% under SLR. Each bank must satisfy the larger of its two equity requirements, so Capital One is constrained by the RBC and maintains equity equal to 10% of risk-weighted assets, while Goldman Sachs is constrained by the SLR and maintains equity equal to 5% of total assets. Capital One therefore faces a 10% marginal equity requirement on credit-card loans, compared with 5% for Goldman Sachs. Goldman Sachs faces a 5% marginal equity requirement on Treasuries, compared with zero for Capital One.

The marginal equity requirement k_{abt}^{Bind} is the partial derivative of the required equity ratio with respect to asset a's portfolio share, holding the other asset shares fixed:

$$\begin{aligned} k_{abt}^{Bind} &= \frac{\partial}{\partial s_{abt}} \max \left\{ \ell_{bt}^{RBC} \sum_a w_a s_{abt},\ \ell_{bt}^{SLR} \sum_a s_{abt} \right\} \\ &= \mathbf{1}\{\text{RBC binds}\} \ell_{bt}^{RBC} w_a + \mathbf{1}\{\text{SLR binds}\} \ell_{bt}^{SLR}. \end{aligned} \tag{2}$$

Financing the required share of asset a with equity adds $c_E k_{abt}^{Bind}$ to funding costs per dollar of the asset, compared with financing the asset entirely with debt.

The asset first-order condition equates marginal net returns to equity funding costs and the opportunity cost of displacing other assets:

$$r'_{abt}(s_{abt}) = c_E k_{abt}^{Bind} + \eta_{bt}. \tag{3}$$

Expanding one asset's portfolio share requires reducing other assets' shares. The multiplier η_{bt} captures the opportunity cost of this displacement, in addition to the direct equity funding cost $c_E k_{abt}^{Bind}$. If the equity funding premium is zero ($c_E = 0$), capital requirements do not affect the bank's asset allocation. If equity is costly, an increase in an asset's equity requirement raises its marginal funding cost. The bank responds by adjusting its portfolio allocation.

2.2 Portfolio Responses to Equity Requirements

An asset's portfolio-share response reflects both its own regulatory cost and substitution induced by changes in other assets' costs. I derive a locally linear relationship between changes in equity requirements and portfolio shares to estimate local own-cost elasticities

[3]This example is adapted from Greenwood et al. (2017).

of asset supply. I assume that r_{abt} is twice continuously differentiable near a fixed portfolio share $\bar{s}_{ab}$, with $r''_{abt}(\bar{s}_{ab}) = -1/\tau_a < 0$. The curvature $1/\tau_a$ is common across banks for each asset and fixed over time. Its inverse, τ_a, measures the asset's local own-cost elasticity of supply. Let Δ denote the change from period $t-1$ to period t, and let $\Delta r'_{abt}(\bar{s}_{ab})$ denote the change in marginal net returns evaluated at the same reference share. Locally approximating marginal net returns and taking differences of the first-order condition gives

$$\Delta s_{abt} \approx \tau_a \left(\Delta r'_{abt}(\bar{s}_{ab}) - c_E \Delta k^{Bind}_{abt} - \Delta \eta_{bt} \right). \tag{4}$$

Appendix A.1 derives equation (4) using a second-order Taylor expansion of net returns around the reference portfolio share $\bar{s}_{ab}$.

The portfolio share s_{abt} responds to changes in per-dollar costs with local supply elasticity τ_a, holding other returns, costs, and the portfolio opportunity cost η_{bt} fixed. The portfolio opportunity cost adjusts when equity requirements change, because the bank must reallocate the displaced portfolio share to other assets. The Lagrange multiplier η_{bt} ensures that portfolio shares sum to one. Holding net-return schedules and bank size fixed, an identical change in the binding equity requirement, k^{Bind}_{abt}, for every asset leaves portfolio shares unchanged.

Proposition 1 shows how the portfolio adding-up identity links each asset's response to its own regulatory cost to its substitution response to changes in other assets' costs.

Proposition 1 (Portfolio substitution). *Suppose net-return schedules and bank size are fixed. Consider small portfolio adjustments around fixed asset shares $\bar{s}_{ab}$. Assume the bank continues to hold a positive amount of every modeled asset and faces one binding capital constraint and one slack capital constraint. The change in an asset's portfolio share in response to changes in required equity across assets is*

$$\Delta s_{abt} \approx -c_E \tau_a \left(\Delta k^{Bind}_{abt} - \Delta \bar{k}^{\tau}_{bt} \right), \qquad \Delta \bar{k}^{\tau}_{bt} = \frac{\sum_a \tau_a \Delta k^{Bind}_{abt}}{\sum_a \tau_a}. \tag{5}$$

Holding net-return schedules and bank size fixed, an asset loses portfolio share when its equity requirement rises by more than the supply-elasticity-weighted average increase in equity requirements across the bank's assets. Since portfolio shares sum to one, an expansion in one asset's portfolio share requires a contraction in other assets' shares. In equation (5), the own-cost component is $-c_E \tau_a \Delta k^{Bind}_{abt}$, and the substitution component is $c_E \tau_a \Delta \bar{k}^{\tau}_{bt}$. Each asset's substitution response depends on the supply elasticities of all assets in the portfolio. The adding-up identity therefore links the own-cost and substitution responses across assets. I use these links to estimate the supply elasticities jointly while accounting for portfolio substitution. Appendix A proves Proposition 1.

2.3 Estimating Own-Cost Supply Elasticities

I estimate each asset's own-cost elasticity of supply, τ_a, from portfolio-share responses to regulatory changes in equity requirements, k_{abt}^{Bind}, calibrating the equity funding premium to $c_E = 5\%$ using estimates in the literature (Appendix C.1). This calibration translates changes in required equity into changes in funding costs.

To allow portfolio shares to adjust over time, I estimate local projections of portfolio-share changes from quarter $t-1$ to quarter $t+h$ on the change in the asset's equity requirement from quarter $t-1$ to quarter t, net of the supply-elasticity-weighted average change across assets:

$$s_{ab,t+h} - s_{ab,t-1} = \delta_{at}^{(h)} + \beta_a \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau}\right) + u_{abt}^{(h)}. \tag{6}$$

Here $\Delta k_{abt}^{Bind} = k_{abt}^{Bind} - k_{ab,t-1}^{Bind}$, and h is the number of quarters after the change in the asset's equity requirement.

Comparing the coefficient on $\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau}$ in the model's portfolio-share response, equation (5), with its coefficient in the estimating equation (6) gives $\beta_a = -c_E \tau_a$. Given the calibrated equity funding premium, the estimated response coefficient therefore recovers the own-cost elasticity of supply:

$$\tau_a = -\frac{\beta_a}{c_E}.$$

Asset a's substitution response, $c_E \tau_a \Delta \bar{k}_{bt}^{\tau}$, depends on its own supply elasticity τ_a and on all assets' supply elasticities through the weights in $\Delta \bar{k}_{bt}^{\tau}$. I therefore estimate all assets' supply elasticities jointly by nonlinear least squares. Although $\Delta \bar{k}_{bt}^{\tau}$ is common across assets held by bank b at date t, the substitution response $c_E \tau_a \Delta \bar{k}_{bt}^{\tau}$ differs across assets when their supply elasticities τ_a differ. Bank–date fixed effects therefore do not generally absorb portfolio substitution. For each candidate set of elasticities, I compute the weighted-average change in required equity, $\Delta \bar{k}_{bt}^{\tau}$, and the predicted share change, $-c_E \tau_a (\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau})$, for each asset. Predicted share changes sum to zero within each bank.

I include asset–date–horizon fixed effects, $\delta_{at}^{(h)}$, to absorb portfolio-share responses to changes in marginal net returns common across banks holding the same asset. These return changes produce a common portfolio-share response across banks because the model assumes a common supply elasticity for each asset. The estimating equation also accommodates changes in marginal net returns common across all assets at a given bank. Such bank–date return changes cancel from portfolio-share responses through the portfolio adding-up identity: raising marginal returns equally across all assets leaves portfolio shares unchanged. Let $\Delta \varepsilon_{abt}$ denote the remaining bank–asset–date return shock after removing the asset–date and bank–date components. I assume that this shock is uncorrelated with regulatory

changes in equity requirements across the bank's assets, conditional on the fixed effects. The residual $u_{abt}^{(h)}$ captures portfolio-share responses to these remaining return shocks and the approximation error from linearizing marginal net returns. Appendix A.2 derives estimating equation (6) from the bank's first-order conditions.

3 Data and Regulatory Variation

3.1 Data and Sample

Data and sample. I use quarterly FR Y-9C data on bank balance sheets and capital ratios for 28 domestic bank holding companies that exceed $100 billion in assets in at least one quarter during the sample.[4] The estimation sample covers 2011–2025.[5] I combine bank-level capital requirements with asset risk weights to measure the equity required per dollar of each asset at each bank and match these requirements to quarterly portfolio shares. I restrict the analysis to assets in a bank's banking book and exclude trading assets from the portfolio-share denominator. I also exclude reserves and vault cash from the assets used to calculate portfolio shares to abstract from surges or contractions in bank reserves driven by quantitative easing or quantitative tightening.

Risk-weight mapping. I match banks' banking-book assets reported on the FR Y-9C to four standardized risk-weight categories—0, 0.2, 0.5, and 1—using the weights prescribed by the U.S. capital rule implementing Basel III (12 CFR Part 217, Subpart D). Table 1 summarizes the asset mapping. I use the labels RW0, RW0.2, RW0.5, and RW1 for these categories, respectively. Appendix Table B.2 lists the FR Y-9C items, assigned risk weights, and legal provisions underlying the mapping. I calculate portfolio shares relative to total banking-book assets, excluding reserves and vault cash. Assets that cannot be assigned to these four risk-weight categories remain in the portfolio-share denominator. The baseline estimation holds their combined portfolio share fixed. Appendix C allows substitution into these remaining assets and estimates their supply elasticity alongside those of the four categories. The estimates for the four categories remain similar to the baseline estimates on the same samples.[6]

[4] I treat banks before and after large mergers (those in which the target exceeds 25% of the acquirer's assets at close) as separate entities so that within-bank changes do not span the acquisitions. Appendix Table B.1 lists the acquisitions. In the estimation below, I cluster standard errors by bank rather than merger-split entity, giving 28 clusters.

[5] The dynamic response estimates use starting quarters from 2011:Q2 through 2024:Q4; the long-run estimates use starting quarters from 2011:Q2 through 2021:Q4. Portfolio outcomes are observed through 2025:Q4, with pre-period outcomes available from 2011:Q1.

[6] See Appendix B.2 for loan coverage and Appendix B.3 for reporting corrections.

Binding rule. A bank's binding rule is the rule that requires the bank to hold the most Tier 1 capital given its portfolio composition and regulatory exposures. Equivalently, it is the rule against which the bank has the smallest Tier 1 capital buffer, measured as actual minus required Tier 1 capital.[7] Before banks began reporting the SLR on the FR Y-9C in 2016:Q3, I use a proxy constructed from FR Y-9C data; Appendix B describes its construction.

I smooth short-lived switches in the binding-rule classification before constructing each asset's required equity. When a bank briefly switches away from a rule and then returns to it, I assign the surrounding rule to the intervening quarters. The procedure generally removes reversals lasting up to four quarters, but preserves spells lasting at least three quarters that begin during the temporary pandemic exemption of reserves and Treasuries from the SLR denominator.[8] As a separate robustness check, Appendix C.13 allows banks to be uncertain about which rule binds and to place continuous weight on both RBC and SLR requirements. I jointly estimate the supply elasticities τ_a and a parameter governing how these weights depend on the difference between the bank's capital buffers under the SLR and RBC requirements. Each buffer is actual Tier 1 capital minus the amount required under the corresponding rule, divided by risk-weighted assets.

3.2 Regulatory Changes and Identifying Variation

I identify asset supply elasticities from two sources of regulatory variation: increases in risk-based capital requirements and switches in banks' binding capital rule. Asset risk weights determine how each change in a bank-level capital requirement translates into changes in the equity required for individual assets. An increase in a binding risk-based capital requirement raises required equity more for assets with higher risk weights. A change in which rule binds can instead raise required equity on some assets while lowering it on others.

Changes to risk-based capital requirements. Changes in the minimum risk-based Tier 1 capital ratio, required Tier 1 capital buffers above that minimum, and bank-specific surcharges generate variation in banks' risk-based capital requirements over the sample period, as shown in Figure 1. Basel III raised the minimum Tier 1 capital requirement from 4% to 6% and introduced a Capital Conservation Buffer for all banks. Additional require-

[7] The Tier 1 leverage requirement occasionally binds instead of the RBC or SLR. I treat it as a separate binding requirement in all estimation. Observations bound by the Tier 1 leverage requirement are excluded from the illustrative figures. Banks also face separate risk-based requirements for common equity Tier 1 (CET1) and total capital; see 12 CFR 217.10. The binding-rule classification used here compares Tier 1 capital requirements and does not separately identify whether the CET1 or total-capital requirement is binding.

[8] Appendix B.3 gives the full smoothing procedure.

ments for the eight largest and most complex banks (G-SIBs) vary over time with their size and systemic importance. The Capital Conservation Buffer and G-SIB surcharges were phased in from 2016 through 2019. Since 2020, banks subject to the Federal Reserve's stress tests must meet a Stress Capital Buffer that varies with each bank's projected losses under stress.[9] I exclude every bank-quarter with a change in the Stress Capital Buffer from the starting events used to estimate asset supply responses. Portfolio-share observations from those quarters remain available as outcomes for other starting events. Section 4.3 uses the 2020 and 2021 Stress Capital Buffer changes to evaluate the estimated supply elasticities.

Changes in banks' binding rule. The introduction of the SLR caused custodian and dealer banks to switch from being constrained by risk-based capital requirements to being constrained by the SLR. Subsequent increases in risk-based requirements and the temporary pandemic SLR exemption caused some banks to switch back. Figure 2 shows that custodians such as State Street and dealers such as Goldman Sachs are usually SLR-bound, while lenders such as Capital One are usually RBC-bound. Custodians and dealers hold large amounts of assets with low risk weights, for which the SLR generally requires more equity than RBC. Lenders hold more high-risk-weight loans, making RBC the tighter requirement. During the pandemic, the temporary exclusion of reserves and Treasuries from the SLR reduced the equity required under the SLR, making even BNY Mellon and State Street RBC-constrained. Goldman Sachs also became RBC-constrained during the later part of the sample as its RBC requirement rose substantially, as shown in Figure 1. A sufficiently large increase in the RBC ratio can make RBC the stricter rule even for a bank whose portfolio is usually SLR-bound.

I compare portfolio-share growth across assets facing different changes in required equity within the same bank, and compare those growth differences across banks holding the same assets. Identification relies on a parallel-trends assumption, as in difference-in-differences. The assumption is that, absent changes in regulation, differences in the portfolio-share growth across assets at the same bank would have been the same as those at other banks.

Because portfolio composition determines which capital rule binds, changes in banks' binding rules can also reflect portfolio adjustments unrelated to regulation. Such endogenous changes in required equity would violate the parallel-trends assumption. For example, an increase in demand for credit-card loans specific to one bank could cause the bank to expand its credit-card holdings relative to other banks and therefore switch from SLR-bound to RBC-bound, raising the equity requirement on its credit-card loans. The bank's increased return on credit-card loans would then both increase its portfolio share in credit cards and raise

[9] I date the increase in the minimum ratio to the Basel III announcement in 2013:Q3. The annual Capital Conservation Buffer increment was 0.625 percentage points, reaching 2.5%. In 2020:Q4, the Stress Capital Buffer replaced that fixed buffer for covered banks, subject to the same 2.5% floor. See the Federal Reserve's 2013 Basel III rule, buffer transition provisions, 2015 G-SIB surcharge rule, and 2020 SCB rule.

required equity on credit cards, biasing my estimated asset supply elasticities toward zero. However, if the bank expands credit-card lending before its equity requirement changes, its credit-card portfolio share would tend to rise relative to other banks before the requirement changes.

Asset-level required equity. I combine each bank's binding capital requirement with asset risk weights to construct the equity required per dollar of each asset, k_{abt}^{Bind}, as defined in equation (2). Expressing the regulatory changes in these common units allows me to combine increases in risk-based requirements and switches in the binding rule in the estimation. For an RBC-bound bank, required equity equals the bank's required RBC ratio multiplied by the asset's risk weight. For an SLR-bound bank, the equity requirement is uniform across assets, except for assets covered by the temporary pandemic exemption. Table 2 reports these asset-specific requirements by bank in 2024:Q4. RBC-bound Capital One faced requirements of 11.5% for credit-card loans, 2.3% for agency mortgage-backed securities, and 0% for Treasuries, while SLR-bound Goldman Sachs faced 5% for every asset.

Figure 3 plots these asset-specific equity requirements over time for a lender, Capital One, and a dealer, Goldman Sachs. Section 2.3 describes how the estimating equation implements these comparisons while accounting for portfolio substitution.

3.3 Empirical Implementation

I pool quarters 16–20 after changes in required equity, when portfolio-share responses begin to plateau for most asset categories (Appendix A.3). I calibrate the equity funding premium to $c_E = 0.05$, implying 5 basis points of additional funding cost per percentage point of required equity. Appendix Table C.1 summarizes the published estimates used for this calibration.

The model assumes a common supply elasticity across banks for each asset. The baseline specification also imposes a common elasticity for RW0 and RW0.2 assets (mostly government securities, interbank lending, and reverse repos), $\tau_0 = \tau_{0.2}$, while estimating separate elasticities for RW0.5 mortgages and RW1 loans. This restriction strengthens identification and improves precision.[10] Appendix Table C.8 reports two alternative methods of pooling assets as robustness checks: one estimates a common elasticity for risk-weight-0 and risk-weight-0.2 assets (mostly government-backed securities and reverse repos) and another for loans (RW0.5 and RW1 assets); the other allows the elasticity to vary linearly with the asset's

[10]Except during the temporary SLR exemption, changes in relative equity requirements are proportional to asset risk weights. These changes generate only three independent portfolio-share responses, because the four responses sum to zero, and therefore cannot separately identify four unrestricted supply elasticities. Pooling the two securities elasticities reduces the number of parameters to three. The temporary exemption provides an additional pattern of relative requirement changes, but unrestricted estimation would rely on that episode to distinguish all four elasticities.

risk weight.

A change in bank-level regulation changes equity requirements on several assets at once. Figure 4 illustrates how an increase in the RBC ratio and a switch in a bank's binding rule both change equity requirements across assets. On the left, for an RBC-bound bank, a one-percentage-point increase in the RBC ratio raises required equity by 1 percentage point for RW1 assets, 0.5 for RW0.5 assets, and 0.2 for RW0.2 assets, while leaving the equity requirement on RW0 assets unchanged. On the right, when a 5% SLR replaces a 10% RBC requirement as the binding constraint, required equity rises on low-risk-weight assets and falls on RW1 assets. In both cases, each asset's portfolio share responds simultaneously to the change in its own equity requirement and changes in other assets' requirements.

4 Estimation Results

I estimate the own-cost elasticities of asset supply using the estimating equation derived in Section 2.3 and the regulatory variation described in Section 3. Correcting for portfolio substitution yields negative long-run portfolio-share responses to required equity for every asset category. The correction changes both the magnitude of the estimated responses and, for mortgages, their sign. The implied own-cost elasticities of asset supply are positive and larger for securities than for loans, with a statistically significant difference between securities and RW1 loans at the 5% level, but not between securities and mortgages.

4.1 Own-Cost Supply Elasticity Estimates

Long-run portfolio-share responses. Figure 5 reports portfolio-share responses to a one-percentage-point increase in required equity, estimated by pooling quarters 16–20. The regression estimates a common response coefficient for each asset across these quarters, with asset–date–horizon fixed effects. The uncorrected specification also includes bank–date fixed effects.[11] Correcting for portfolio substitution makes the estimated response more negative for every asset and reverses the mortgage coefficient from positive to negative. Without the correction, the mortgage coefficient has the wrong sign: it implies that higher equity requirements increase mortgages' portfolio share. The response of risk-weight-1 assets remains negative and changes relatively little with the correction. Appendix Table C.8 reports the corresponding supply elasticities and standard errors for the plotted estimates

[11] Pooling estimates a single long-run response from the underlying observations rather than averaging separately estimated coefficients for each quarter. Asset–date–horizon fixed effects allow each asset's average response across banks to differ for every starting date and horizon.

and alternative specifications; multiplying them by $-c_E$ gives the response coefficients and multiplying standard errors by c_E gives their standard errors.

The upward bias in the uncorrected regression's coefficients on required equity arises from switches between RBC and SLR in the sample. The uncorrected regression includes bank–date fixed effects and compares portfolio-share growth across assets within the same bank. During switches between RBC and SLR, securities' higher supply elasticities make their portfolio shares respond more strongly than loan shares to the common substitution term. Substitution therefore narrows the difference in growth between securities and loans compared with what the same elasticities predict from changes in each asset's own equity requirement alone. Without controlling for substitution, the regression attributes this narrowing to weaker responses of portfolio shares to required equity. Neglecting substitution in response to other regulatory changes could instead bias the coefficients downward. Both the model's prediction and the estimated correction imply more negative portfolio-share responses to required equity in every asset category.

Adjustment over time. Figure 6 traces the portfolio-share responses over the quarters following a change in required equity. The substitution-corrected estimates show how the negative long-run responses develop over time. For most asset categories, the responses flatten over quarters 16–20, supporting the interpretation of the pooled estimates as long-run responses.

Implied supply elasticities. I convert the long-run response coefficients into supply elasticities using $\widehat{\tau}_a = -\widehat{\beta}_a/c_E$ and the equity funding premium $c_E = 0.05$ calibrated in Section 3.3. Figure 7 and Table 3 report the resulting estimates. The implied supply elasticities are larger for securities than for loans. The difference between securities and RW1 loans is statistically significant at the 5% level ($p = 0.031$); the difference between securities and mortgages is not ($p = 0.118$) (Appendix Table C.2).

Table 3 compares supply elasticities estimated without portfolio-substitution correction with estimates that include the correction.

Implied portfolio response when the SLR becomes binding. Consider a bank switching from a binding 10% RBC requirement to a binding 5% SLR requirement, holding net-return schedules and total assets fixed. Under RBC, required equity equals the 10% bank-level requirement multiplied by the asset's risk weight; under SLR, required equity is 5% for every asset. Using the risk weights in Table 1, required equity changes by 5, 3, 0, and -5 percentage points for RW0, RW0.2, RW0.5, and RW1, respectively. The substitution-corrected elasticities in Table 3 are 30.5 for each securities category, 11.6 for mortgages, and 4.4 for RW1 assets. These elasticities weight the changes in assets' equity requirements in

the portfolio-substitution benchmark:

$$\Delta\bar{k}^{\tau} = \frac{(30.5 \times 5) + (30.5 \times 3) + (11.6 \times 0) + (4.4 \times (-5))}{30.5 + 30.5 + 11.6 + 4.4} \approx 2.9.$$

For government-backed securities (RW0), subtracting the average change in equity requirements across the bank's assets gives an increase above that average of $5-2.9 = 2.1$ percentage points. Applying the calibrated equity premium, $c_E = 5\%$ (Appendix Table C.1), and the RW0 elasticity gives

$$\Delta s_0 \approx -\frac{5}{100} \times 30.5 \times (5 - 2.9) \approx -3.2 \text{ percentage points.}$$

For C&I, credit cards, and other unsecured loans (RW1), the change in their equity requirement minus the average change across the bank's assets is $-5-2.9 = -7.9$ percentage points. Using the RW1 elasticity gives

$$\Delta s_1 \approx -\frac{5}{100} \times 4.4 \times (-5 - 2.9) \approx +1.7 \text{ percentage points.}$$

In this switch from a binding 10% RBC requirement to a binding 5% SLR requirement, mortgages gain portfolio share through substitution: their equity requirement remains at 5%, while the average change in equity requirements across the bank's assets is positive.

4.2 Alternative Specifications

Appendix C conducts robustness checks of the asset supply elasticity estimates, retaining the baseline assumption of a single binding capital rule. Appendix C.13 estimates the weights banks place on RBC and SLR requirements jointly with asset supply elasticities.

One concern with the estimation is that banks may change their portfolios for reasons unrelated to regulation and thereby change their binding capital rules. Measured changes in required equity would then partly reflect banks' endogenous portfolio choices. I test for portfolio-share trends before changes in required equity and find no statistically significant pre-event coefficients (Figure C.1). To account for portfolio trends that began before the regulatory event, I separately control for lagged portfolio shares (Table C.3). In a separate specification, I recalculate changes in required equity to exclude changes caused solely by banks' contemporaneous portfolio choices. To do this, I apply each bank's contemporaneous regulatory ratios to its previous-quarter portfolio composition to determine the binding rule (Table C.4). The resulting supply elasticity estimates are smaller than but qualitatively similar to the baseline estimates.

Banks may respond differently to temporary and permanent changes in capital requirements. The Fed's emergency exemption of reserves and Treasuries from the SLR over 2020–2021 was temporary and coincided with unusual economic conditions. I reestimate the supply elasticities excluding the introduction and expiration of the exemption from the set of regulatory events. The resulting estimates are larger and less precise than the baseline estimates (Figure C.2 and Table C.5).

Portfolio-share changes measured over 4–5 years can reflect responses to regulatory changes before or after the event quarter used in the regression. I control for changes in required equity before and after that event quarter to separate their effects from the response to required equity in the event quarter (Table C.6). Adding these controls yields point estimates similar to the baseline estimates, so I use the baseline estimates in the policy application exercise of Section 5.

The baseline standard errors are clustered by bank. I report standard errors clustered by both bank and quarter to allow cross-bank correlation. Conventional clustered tests may be inaccurate with only 28 bank clusters. I therefore report wild cluster bootstrap inference alongside the conventional tests (Table C.7).

The baseline allows separate elasticities for RW0.5 and RW1 loans while pooling RW0 and RW0.2 securities. In Table C.8, I instead impose a common elasticity for RW0.5 and RW1 loans, or allow elasticities to vary linearly with risk weight. The estimated supply elasticities remain positive and larger for securities than for loans.

The baseline specification holds fixed the portfolio share of banking-book assets outside the four modeled categories. I relax this restriction by estimating a supply elasticity for those assets as a fifth category in Table C.9. The elasticity estimates for the four modeled categories are similar to the baseline estimates reestimated on the same observations.

To assess sensitivity to the measurement of asset risk weights, I compare supply elasticity estimates based on the standardized risk-weight mapping with estimates based on banks' reported allocations of assets to each risk-weight category in Schedule HC-R, the regulatory capital schedule of the FR Y-9C. The detailed breakdown is available from 2015 onward, so I restrict the comparison to regulatory changes from 2015 onward and use the same bank observations for both specifications (Table C.10). The resulting supply elasticity estimates are larger than but qualitatively similar to the baseline estimates on the same sample.

I use an annual funding premium for equity relative to debt of 5 percentage points ($c_E = 0.05$) to recover supply elasticities from estimated portfolio-share responses. If the true equity funding premium exceeds 5 percentage points, each percentage-point increase in required equity raises funding costs by more than the baseline calibration assumes. The same estimated portfolio-share response would then imply a smaller supply elasticity than

the baseline estimate. Likewise, if the actual equity premium is below 5 percentage points, that would imply a larger supply elasticity than the baseline estimate (Figure C.4).

In practice, banks likely place nonzero weight on both the RBC and SLR requirements. Appendix C.13 estimates the parameter governing those weights jointly with the supply elasticities. The resulting supply elasticity estimates remain positive and larger for securities than for loans, but are less precise than the baseline estimates.

4.3 Out-of-Sample Validation

The model predicts the direction of portfolio-share responses to the 2020 and 2021 Stress Capital Buffer (SCB) changes in three of the four asset categories. Bank-quarters with SCB changes are excluded as starting events from the baseline estimation. I evaluate the baseline estimates by comparing their predictions with responses estimated using SCB changes alone.[12]

Regulatory changes used for validation. The Stress Capital Buffer is a bank-specific risk-based capital buffer determined by supervisory stress-test losses and planned dividends, subject to a minimum of 2.5%. Figure 8 shows changes in required buffers associated with its introduction in 2020:Q4 and update in 2021:Q4. These changes provide a separate source of regulatory variation with which to evaluate the estimated supply elasticities. I hold other regulatory requirements and pre-event regulatory ratios fixed, change the capital buffer, and recompute each bank's binding rule and required equity across assets. The resulting predictions incorporate both changes in each asset's own required equity and substitution across the bank's portfolio.[13]

Predicted vs. estimated responses of portfolio shares to SCB. Figure 9 compares two predictions for the same SCB-induced changes in required equity: one uses the baseline supply elasticities, and the other uses elasticities estimated from SCB changes alone. Both incorporate portfolio substitution and average responses over the same bank–event–quarter observations with a nonzero buffer change, pooling available quarters between 16 and 20 after each event. Banks without buffer changes remain in the regression comparison group. Outcomes through 2025:Q4 provide horizons 16–20 for the 2020 event and horizon 16 for the 2021 event.

[12]Portfolio-share observations from SCB-change quarters remain available as outcomes for other starting events. The validation therefore uses excluded regulatory events, while its outcome periods can overlap with those used in the baseline estimation.

[13]The calculation retains the temporary SLR exemption in effect at the 2020 event. For Morgan Stanley, the exemption reduces the SLR charge on the exempt portion of RW0 holdings, so the SCB-induced changes in assets' equity requirements are not an affine function of risk weights across all four categories. The resulting RW0.2 prediction therefore need not be a fixed fraction of the RW0 prediction based solely on the baseline elasticities and risk weights.

The baseline model predicts a 0.7-percentage-point increase in the portfolio share of government-backed securities (RW0), compared with a response estimated using SCB changes alone of 1.0 percentage points. For agency securities and interbank assets (RW0.2), the baseline model predicts a 0.1-percentage-point increase, while the estimate using SCB changes alone is a 0.2-percentage-point decline. Both sets of estimates imply a decline in mortgage shares: 0.4 percentage points under the baseline model and 0.6 percentage points using SCB changes alone. For RW1 assets, the corresponding declines are 0.4 and 0.2 percentage points.

Estimates using SCB changes alone reproduce the predicted direction of portfolio reallocation in three of the four asset categories; the RW0.2 estimate has the opposite sign. Their confidence intervals include the baseline predictions in all four categories, although the estimates are too imprecise to distinguish the predicted magnitudes sharply.

4.4 Model-Implied Comparative Advantage

Beyond these portfolio responses, I use the estimated supply elasticities to infer differences in banks' net returns that account for their persistent specialization after controlling for equity requirements. I use data on banks' portfolio shares in each asset to recover bank-specific net-return levels on each asset, using the estimated supply elasticities to convert cross-bank differences in portfolio shares into implied differences in net returns. The estimates reveal comparative advantages in lending for traditional banks and in securities and reverse repos for dealers and custodians, each worth several percentage points in annual net returns.

To recover bank-specific net-return levels, I assume that each asset's marginal net return declines at a constant rate across portfolio allocations. Under this assumption, a bank holding a larger share of an asset despite a higher equity requirement must have a higher net-return intercept for that asset, holding the bank's portfolio opportunity cost fixed. I specify $r_{abt}(s) = (\bar{r}_{at} + \bar{r}_{ab} + \varepsilon_{abt})s - s^2/(2\tau_a)$, where $\bar{r}_{at} + \bar{r}_{ab} + \varepsilon_{abt}$ is the intercept of marginal net returns. The common asset–date net-return level $\bar{r}_{at}$ captures variation in asset a's net-return level shared by all banks. The intercept $\bar{r}_{ab}$ is the persistent bank-specific component of the net-return level on asset a. Differences in these intercepts across banks capture comparative advantages in supplying the asset. I assume that the remaining net-return shock ε_{abt} has zero time mean for each bank–asset pair. Substituting the quadratic net-return function and return decomposition into the bank's first-order condition gives

$$\frac{s_{abt}}{\widehat{\tau}_a} + c_E k_{abt}^{Bind} = \bar{r}_{at} + \bar{r}_{ab} - \eta_{bt} + \varepsilon_{abt},$$

where η_{bt} is the bank's multiplier on the portfolio adding-up identity.

I recover the net-return intercepts $\bar{r}_{ab}$ by estimating bank–asset fixed effects in a regression of $s_{abt}/\widehat{\tau}_a + c_E k_{abt}^{Bind}$ on bank–asset, asset–date, and bank–date fixed effects. Asset–date fixed effects absorb common net-return levels, while bank–date fixed effects absorb the multiplier on the portfolio adding-up identity. I normalize the intercepts to average zero across banks within each asset and across assets within each bank. Confidence intervals use Newey–West standard errors with four quarterly lags, allowing regression errors to be correlated across observations within the same calendar quarter and across quarters up to four quarters apart. They treat the estimated supply elasticities and equity funding premium as fixed. A positive intercept indicates a comparatively high net-return level on that asset after removing bank-wide and asset-wide net-return differences.

The estimated comparative advantages align with banks' business models: Goldman Sachs and Morgan Stanley have estimated net-return intercepts of about 3 percentage points per year in RW0 assets, which for these banks are predominantly made up of reverse repos, while custodians State Street and BNY Mellon have intercepts of about 2.5 percentage points per year in both RW0 and RW0.2 assets (Figure 10). Credit-card lenders American Express and Capital One have estimated net-return intercepts in RW1 loans of about 8 and 3 percentage points per year, respectively.

5 Application to Post-Crisis Capital Requirements

Since the financial crisis, large U.S. banks have faced stronger capital requirements while simultaneously shifting their portfolios away from lending and toward securities (Hanson et al., 2024). A prominent explanation is that stronger bank regulation has discouraged bank lending and encouraged lending by nonbanks (Buchak et al., 2018; Irani et al., 2021; Lee et al., 2024). Other work emphasizes technological change and shifts in borrower and saver demand (Buchak et al., 2024). I use the estimated asset supply elasticities to quantify the contribution of capital requirements to this shift and to the convergence in banks' portfolio compositions.

5.1 Aggregate portfolio composition

Post-crisis regulatory reforms raised risk-based capital requirements and introduced the SLR, changing equity requirements for different assets across banks. Figure 11 summarizes these regulatory changes. Higher buffers and bank-specific surcharges increased RBC requirements, while the SLR became the binding requirement for banks whose portfolios were concentrated in low-risk-weight assets.

For each bank and quarter, I determine the binding capital rule and calculate the equity required for each asset. I compare these requirements with a counterfactual that retains each bank's average RBC requirement over 2011:Q1–2013:Q2, before the SLR was announced. I apply the estimated supply elasticities to approximate the resulting portfolio changes, accounting for portfolio substitution as in Proposition 1. Predicted and observed changes use the same available quarters within each bank and fixed bank aggregation weights. The calculation holds bank sizes and net-return schedules fixed.

My estimates predict that the post-crisis strengthening of bank capital requirements induced banks to reallocate their portfolios from lending toward securities, as observed in the data. Figure 12 compares predicted and observed changes in large banks' portfolio composition between 2011–2013 and 2022–2024. Table 4 reports actual portfolio-share changes, model-implied changes due to capital requirements, and the model-implied changes as percentages of the actual changes. The model explains about 28% of the shift from lending toward securities between 2011–2013 and 2022–2024, predicting an increase in securities' portfolio share minus loans' portfolio share of 3.9 percentage points compared with the observed increase of 14.0 percentage points. Table 4 reports confidence intervals for the model-implied percentage-point changes.[14]

The SLR partly offsets large banks' aggregate shift from lending to securities induced by stronger risk-based capital requirements. Higher RBC requirements raise the equity cost of high-risk-weight loans relative to securities. When the SLR becomes binding instead of the RBC, it raises required equity on low-risk-weight securities and lowers it on loans relative to the RBC requirement. The model predicts that, absent the SLR, there would have been a larger shift from lending to securities (Figure 12).

Stronger bank capital requirements therefore account for part, but not most, of large banks' shift from lending to securities over 2011–2024. Other forces, including changes in nonbank lending technology, may also have contributed (Hanson et al., 2024; Buchak et al., 2024). In general equilibrium, an aggregate strengthening of capital requirements is likely to raise loan rates as well as reduce loan quantities, so banks may shift their portfolios away from lending less than this partial-equilibrium exercise predicts. My cross-sectional estimates may also capture borrower substitution from banks facing larger capital requirement increases toward those facing smaller increases. Applying these estimates to a broad strengthening of capital requirements could therefore overstate the aggregate portfolio shift away from lending, because fewer banks remain available to absorb displaced borrowers than when

[14]The model explains a larger share of the increase in securities shares (37%) than of the decline in loan shares (23%) because the observed increase in securities shares is smaller than the observed decline in loan shares, whereas the model predicts equal and opposite changes.

requirements rise at only some banks.

5.2 Portfolio convergence across banks

Post-crisis capital requirements may also have contributed to the convergence in large banks' average asset risk weights since 2011. Since each bank tends to be bound by the rule that most strongly penalizes its core assets, risk-based capital requirements encourage lending specialists to expand securities holdings, while the leverage requirement encourages dealers and custodians to expand lending (Greenwood et al., 2017). These incentives can make banks' portfolios more similar in regulatory risk weight. I use the estimated supply elasticities to quantify how much of the observed convergence capital requirements can explain.

Consider again the two-bank example in Section 2, with Capital One specializing in credit-card loans and Goldman Sachs specializing in Treasuries. With a 10% RBC requirement and a 5% SLR, Capital One is RBC-bound and Goldman Sachs is SLR-bound. Capital One must fund each additional $100 of credit-card loans with $10 of equity, compared with $5 for Goldman Sachs. Conversely, Goldman Sachs must fund each additional $100 of Treasuries with $5 of equity, while Capital One faces no equity requirement on Treasuries. Each bank therefore faces a higher equity requirement on its specialty than the other bank does, encouraging both to expand outside their specialties.

I measure the bank's average risk weight across the four asset categories, weighted by each category's share of the assets in those categories. Convergence means that banks with higher initial average risk weights subsequently reduced them more—or increased them less—than banks with lower initial average risk weights.

Banks' ratios of risk-weighted assets to assets converged between 2011–2013 and 2022–2024, consistent with the earlier evidence in Greenwood et al. (2017). Figure 13 plots changes in RWA/assets against banks' initial RWA/assets. Goldman Sachs and Morgan Stanley began with low RWA/assets and subsequently increased their ratios. Wells Fargo, PNC, and U.S. Bancorp began with higher RWA/assets and subsequently reduced their ratios. These portfolio shifts are consistent with the incentives of SLR-bound dealers to expand lending and RBC-bound lenders to expand their securities portfolios. Regressing the observed change in RWA/assets on initial RWA/assets gives a slope of -0.14. Regressing the model-implied change on the same initial RWA/assets gives a slope of -0.05. The ratio of the model-implied slope to the observed slope implies that capital requirements account for about 35% of the observed convergence. Table D.1 reports confidence intervals for both slopes.

6 Conclusion

Capital requirements affect banks' portfolio composition both by changing each asset's own equity requirement and by changing equity requirements on other assets. This paper develops a portfolio-allocation model and empirical method to estimate each asset's own-cost elasticity of supply using changes in bank-level capital requirements that alter equity requirements for multiple assets simultaneously. In the model, each asset's own-cost elasticity governs both its direct response to its own regulatory cost and its substitution response to changes in other assets' regulatory costs. The estimates imply that raising an asset's equity requirement reduces its portfolio share for every asset risk-weight category, holding other assets' equity requirements fixed, with securities responding more strongly than loans.

Applying these estimates to post-crisis capital requirements, I find that stronger capital regulation accounts for about 28% of large banks' shift from lending toward securities since 2011 and 35% of the convergence in their average asset risk weights. These changes reflect different regulatory incentives across banks with different business models: risk-based capital requirements encourage lending specialists to expand securities holdings, while the leverage requirement encourages dealers and custodians to expand lending. Capital regulation thus contributes to the aggregate decline in lending shares while also encouraging banks to expand outside their specialties.

These results show that the effect of capital regulation on banks' portfolio composition depends on how requirements change across the entire portfolio. Raising equity requirements on securities can encourage lending even without reducing the equity required to fund loans. Raising requirements on loans can instead encourage substitution toward securities. Evaluating a new policy therefore requires considering both its effects on the assets it directly targets and the reallocation it induces elsewhere in banks' portfolios. The framework developed here quantifies these portfolio adjustments, holding bank sizes fixed; assessing their implications for total credit provision also requires accounting for changes in bank size and lending outside the banking sector.

References

Acharya, V. V., N. Cetorelli, and B. Tuckman (2026, August). Where Do Banks End and NBFIs Begin? *The Review of Corporate Finance Studies 15*(3), 627–659.

Aiyar, S., C. W. Calomiris, and T. Wieladek (2014, February). Does Macro-Prudential Regulation Leak? Evidence from a UK Policy Experiment. *Journal of Money, Credit and Banking 46*(s1), 181–214.

Basel Committee on Banking Supervision (2010, August). An assessment of the long-term economic impact of stronger capital and liquidity requirements. Technical Report BCBS 173.

Begenau, J. and T. Landvoigt (2022, July). Financial Regulation in a Quantitative Model of the Modern Banking System. *The Review of Economic Studies 89*(4), 1748–1784.

Behn, M., R. Haselmann, and P. Wachtel (2016, April). Procyclical Capital Regulation and Lending. *The Journal of Finance 71*(2), 919–956.

Benetton, M. (2021, December). Leverage Regulation and Market Structure: A Structural Model of the U.K. Mortgage Market. *The Journal of Finance 76*(6), 2997–3053.

Benetton, M., P. Eckley, N. Garbarino, L. Kirwin, and G. Latsi (2021, October). Capital requirements and mortgage pricing: Evidence from Basel II. *Journal of Financial Intermediation 48*, 100883.

Berrospide, J. M. and R. M. Edge (2019, July). The Effects of Bank Capital Buffers on Bank Lending and Firm Activity: What Can We Learn from Five Years of Stress-Test Results? *Finance and Economics Discussion Series 2019.0*(50).

Blattner, L., L. Farinha, and F. Rebelo (2023, June). When Losses Turn into Loans: The Cost of Weak Banks. *American Economic Review 113*(6), 1600–1641.

Brainard, W. C. and J. Tobin (1968). Pitfalls in Financial Model Building. *American Economic Review 58*(2), 99–122.

Bridges, J., D. Gregory, M. Nielsen, S. Pezzini, A. Radia, and M. Spaltro (2014). The Impact of Capital Requirements on Bank Lending. *SSRN Electronic Journal*.

Brooke, M., O. Bush, R. Edwards, J. Ellis, B. Francis, R. Harimohan, K. Neiss, and C. Siegert (2015). Measuring the macroeconomic costs and benefits of higher UK bank capital requirements. Financial Stability Paper 35, Bank of England.

Bräuning, F. and H. Stein (2024, July). The Effect of Primary Dealer Constraints on Intermediation in the Treasury Market. Federal Reserve Bank of Boston Research Department Working Papers, Federal Reserve Bank of Boston. Series: Federal Reserve Bank of Boston Research Department Working Papers.

Buchak, G., G. Matvos, T. Piskorski, and A. Seru (2018, December). Fintech, regulatory arbitrage, and the rise of shadow banks. *Journal of Financial Economics 130*(3), 453–483.

Buchak, G., G. Matvos, T. Piskorski, and A. Seru (2024, February). The Secular Decline of Bank Balance Sheet Lending. Technical Report w32176, National Bureau of Economic Research, Cambridge, MA.

Chernenko, S., R. Ialenti, and D. S. Scharfstein (2025). Bank Capital and the Growth of Private Credit.

DeAngelo, H. and R. M. Stulz (2015, May). Liquid-claim production, risk management, and bank capital structure: Why high leverage is optimal for banks. *Journal of Financial Economics 116*(2), 219–236.

Deaton, A. and J. Muellbauer (1980, June). An Almost Ideal Demand System. *The American Economic Review 70*(3), 312–326.

Diamond, W., Z. Jiang, and Y. Ma (2024, September). The reserve supply channel of unconventional monetary policy. *Journal of Financial Economics 159*, 103887.

Dietsch, M., H. Fraisse, M. Lé, and S. Lecarpentier (2020). Lower Bank Capital Requirements as a Policy Tool to Support Credit to SMEs: Evidence From a Policy Experiment? *SSRN Electronic Journal*.

Du, W., A. Tepper, and A. Verdelhan (2018, June). Deviations from Covered Interest Rate Parity. *The Journal of Finance 73*(3), 915–957.

Duffie, D., M. J. Fleming, F. M. Keane, C. Nelson, O. Shachar, and P. Van Tassel (2023, August). Dealer Capacity and U.S. Treasury Market Functionality. Staff Reports (Federal Reserve Bank of New York), Federal Reserve Bank of New York. Series: Staff Reports (Federal Reserve Bank of New York).

Duffie, D. and A. Krishnamurthy (2016). Pass-Through Efficiency in the Fed's New Monetary Policy Setting.

Favara, G., S. Infante, and M. Rezende (2022). Leverage Regulations and Treasury Market Participation: Evidence from Credit Line Drawdowns. *SSRN Electronic Journal*.

Favara, G., I. Ivanov, and M. Rezende (2021, December). GSIB surcharges and bank lending: Evidence from US corporate loan data. *Journal of Financial Economics 142*(3), 1426–1443.

Firestone, S., A. Lorenc, and B. Ranish (2017, March). An Empirical Economic Assessment of the Costs and Benefits of Bank Capital in the US. *Finance and Economics Discussion Series 2017.0*(34).

Fraisse, H., M. Lé, and D. Thesmar (2020, January). The Real Effects of Bank Capital Requirements. *Management Science 66*(1), 5–23.

Fuchs, W. M., S. Fukuda, and D. Neuhann (2023). Demand-System Asset Pricing: Theoretical Foundations.

Furfine, C. (2001). Bank Portfolio Allocation: The Impact of Capital Requirements, Regulatory Monitoring, and Economic Conditions. *Journal of Financial Services Research 20*(1), 33–56.

Gorton, G. and G. Pennacchi (1990). Financial Intermediaries and Liquidity Creation. *The Journal of Finance 45*(1), 49–71.

Gorton, G. and A. Winton (2017, February). Liquidity Provision, Bank Capital, and the Macroeconomy. *Journal of Money, Credit and Banking 49*(1), 5–37.

Greenwood, R., J. C. Stein, S. G. Hanson, and A. Sunderam (2017). Strengthening and Streamlining Bank Capital Regulation. *Brookings Papers on Economic Activity 2017*(2), 479–565.

Gropp, R., T. Mosk, S. Ongena, and C. Wix (2019, January). Banks Response to Higher Capital Requirements: Evidence from a Quasi-Natural Experiment. *The Review of Financial Studies 32*(1), 266–299.

Haddad, V., Z. He, P. Huebner, P. Kondor, and E. Loualiche (2026). Causal Inference for Asset Pricing.

Hanson, S. G., V. Ivashina, L. Nicolae, J. C. Stein, A. Sunderam, and D. K. Tarullo (2024, March). The Evolution of Banking in the 21st Century: Evidence and Regulatory Implications. *Brookings Papers on Economic Activity 2004*(1), 343–389.

He, Z., S. Nagel, and Z. Song (2022, January). Treasury inconvenience yields during the COVID-19 crisis. *Journal of Financial Economics 143*(1), 57–79.

Irani, R. M., R. Iyer, R. R. Meisenzahl, and J.-L. Peydró (2021, April). The Rise of Shadow Banking: Evidence from Capital Regulation. *The Review of Financial Studies 34*(5), 2181–2235.

Juelsrud, R. E. and E. G. Wold (2020, January). Risk-weighted capital requirements and portfolio rebalancing. *Journal of Financial Intermediation 41*, 100806.

Kashyap, A. K., J. C. Stein, and S. G. Hanson (2010). An Analysis of the Impact of "Substantially Heightened" Capital Requirements on Large Financial Institutions. Working paper, University of Chicago Booth School of Business and Harvard University.

Koijen, R. S. J. and M. Yogo (2019). A Demand System Approach to Asset Pricing. *Journal of Political Economy 127*(4), 1475–1515.

Koont, N. and S. Walz (2026, July). The Declining Role of Deposits in Credit Creation.

Lee, H., S. Lee, and R. Paluszynski (2024, September). Capital Regulation and Shadow Finance: A Quantitative Analysis. *Review of Economic Studies 91*(5), 3047–3084.

Li, J., Y. Ma, C. Mendicino, and D. Supera (2026, April). Bank to non-bank lending and the reallocation of credit. Technical Report 978-92-899-7854-5, European Central Bank, Frankfurt am Main. 3220.

Mayordomo, S. and M. Rodríguez-Moreno (2018, October). Did the bank capital relief induced by the Supporting Factor enhance SME lending? *Journal of Financial Intermediation 36*, 45–57.

Miles, D., J. Yang, and G. Marcheggiano (2013, March). Optimal Bank Capital. *The Economic Journal 123*(567), 1–37.

Plosser, M. C. and J. A. C. Santos (2024, February). The Cost of Bank Regulatory Capital. *The Review of Financial Studies 37*(3), 685–726.

Stein, J. C. (2012, February). Monetary Policy as Financial Stability Regulation*. *The Quarterly Journal of Economics 127*(1), 57–95.

Sundaresan, S. and K. Xiao (2024). Liquidity regulation and banks: Theory and evidence. *Journal of Financial Economics 151*, 103747.

Wang, Y., T. M. Whited, Y. Wu, and K. Xiao (2022). Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation. *The Journal of Finance 77*(4), 2093–2141.

Tables and Figures

Risk weight	Examples of assets
0	U.S. Treasuries; GNMA mortgage-backed securities
0.2	Federal funds; agency MBS; loans to U.S. banks
0.5	First-lien residential mortgages
1	C&I; credit cards; commercial real estate; other consumer loans

Table 1: Asset categories and standardized risk weights. [Back to text]

Notes: MBS denotes mortgage-backed securities; GNMA denotes the Government National Mortgage Association; C&I denotes commercial and industrial loans. Appendix Table B.2 lists the FR Y-9C reporting fields used to measure each asset category and the regulatory provisions assigning its risk weight. *Sources:* FR Y-9C reporting instructions; 12 CFR Part 217.

	Binding rule	C&I loans RW1	Credit cards RW1	Residential mortgages RW0.5	Agency MBS RW0.2	Treasuries RW0
G-SIBs						
BNY Mellon	SLR	5	5	5	5	5
Goldman Sachs	SLR	5	5	5	5	5
Morgan Stanley	SLR	5	5	5	5	5
State Street	SLR	5	5	5	5	5
Bank of America	RBC	12.2	12.2	6.1	2.4	0
Citigroup	RBC	13.6	13.6	6.8	2.7	0
JPMorgan Chase	RBC	13.8	13.8	6.9	2.8	0
Wells Fargo	RBC	11.3	11.3	5.7	2.3	0
Other large bank holding companies						
Capital One	RBC	11.5	11.5	5.8	2.3	0
PNC Financial	RBC	8.5	8.5	4.2	1.7	0
Truist Financial	RBC	8.8	8.8	4.4	1.8	0
U.S. Bancorp	RBC	9.1	9.1	4.6	1.8	0

Table 2: Required equity by bank and asset, 2024:Q4 (percent). [Back to text]

Notes: Required equity is the percentage of funding for each asset that must come from equity (Tier 1 capital) under the bank's binding capital rule. RBC denotes risk-based capital; SLR denotes the supplementary leverage ratio. RW denotes standardized risk weight; C&I denotes commercial and industrial loans; MBS denotes mortgage-backed securities; G-SIB denotes a global systemically important bank. For RBC-bound banks, required equity equals the required Tier 1 capital ratio times the standardized asset risk weight. For SLR-bound banks, the required equity funding share equals the bank's required SLR ratio. The bank's binding rule is the rule that requires the most Tier 1 capital given its portfolio composition and regulatory exposures. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

	No substitution correction	Pooled RW0/0.2 with substitution correction
RW0	16.65**	30.49***
	(7.36)	(10.99)
RW0.2	8.21	30.49***
	(7.40)	(10.99)
RW0.5	−12.19	11.62*
	(13.67)	(5.83)
RW1	2.50	4.41*
	(3.79)	(2.48)
N. Banks	28	28
N. Quarters	42	42
SE clustering	Bank	Bank
Adjusted R^2	0.13	0.11
Adjusted within R^2	0.00	0.00
Bank–date FE	Yes	No
Asset–date–horizon FE	Yes	Yes

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Long-run supply elasticities with and without portfolio-substitution correction. [Back to text]

Notes: RW denotes the standardized risk weight. The corrected specification imposes a common elasticity for RW0 and RW0.2 and separate elasticities for RW0.5 and RW1; the uncorrected specification estimates all four separately. Estimates pool quarters 16–20. Elasticities are recovered as $\widehat{\tau}_a = -\widehat{\beta}_a / c_E$, where $\widehat{\beta}_a$ is the portfolio-share response to required equity and $c_E = 0.05$ is the equity funding premium. Inference treats c_E as fixed. *Sources:* FR Y-9C bank regulatory filings.

	Observed change (percentage points)	Model-implied change (percentage points)	Percentage explained
RW0	4.8	1.9 [0.8, 2.9]	38.7%
RW0.2	0.5	0.1 [-0.7, 0.9]	21.5%
RW0.5	-2.3	-1.0 [-1.8, -0.1]	42.4%
RW1	-6.4	-1.0 [-2.1, 0.1]	15.6%
Securities minus loans	14.0	3.9 [1.1, 6.8]	28.2%

Table 4: Observed and model-implied changes in aggregate portfolio shares. [Back to text]

Notes: Changes compare 2011:Q1–2013:Q2 with 2022–2024. Bank weights are fixed at pre-period averages. Observed and model-implied changes use the same available quarters within each bank and comparison period. Predictions hold bank sizes and net-return schedules fixed. Brackets report 95% confidence intervals reflecting uncertainty in the bank-clustered supply-elasticity estimates. The final row subtracts the change in RW0.5 and RW1 shares from the change in RW0 and RW0.2 shares; its interval accounts jointly for estimation uncertainty across all four categories. Percentages are point estimates only. Shares exclude trading assets, reserves, and vault cash. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

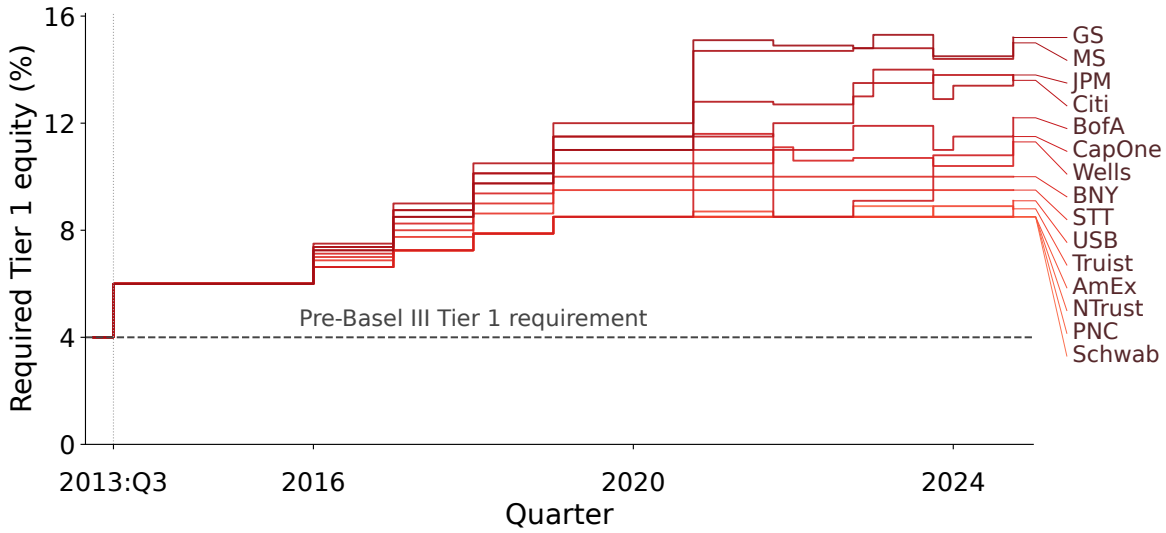


Figure 1: Bank-level RBC requirements over time. [Back to text]

Notes: Each line shows a bank's required Tier 1 capital as a percentage of risk-weighted assets, including the minimum ratio, buffers, and surcharges. The dashed horizontal line marks the pre-Basel III minimum. RBC denotes risk-based capital. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules.

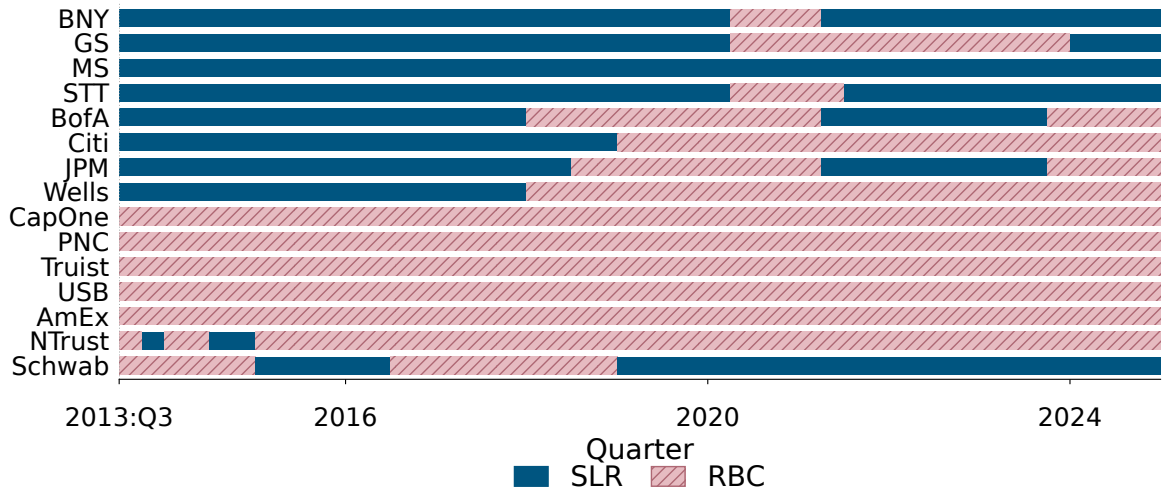


Figure 2: Binding capital rule by bank. [Back to text]

Notes: Each row represents a bank. Hatched light-red segments indicate a binding risk-based capital (RBC) requirement; solid blue segments indicate a binding supplementary leverage ratio (SLR) requirement. The bank's binding rule is the rule that requires the most Tier 1 capital given its portfolio composition and regulatory exposures. The figure identifies the binding rule by comparing RBC and SLR requirements, excluding the separate Tier 1 leverage requirement. The estimation includes the separate Tier 1 leverage requirement, which rarely binds. Before 2016:Q3, I approximate the denominator of the SLR using the method described in Appendix B. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

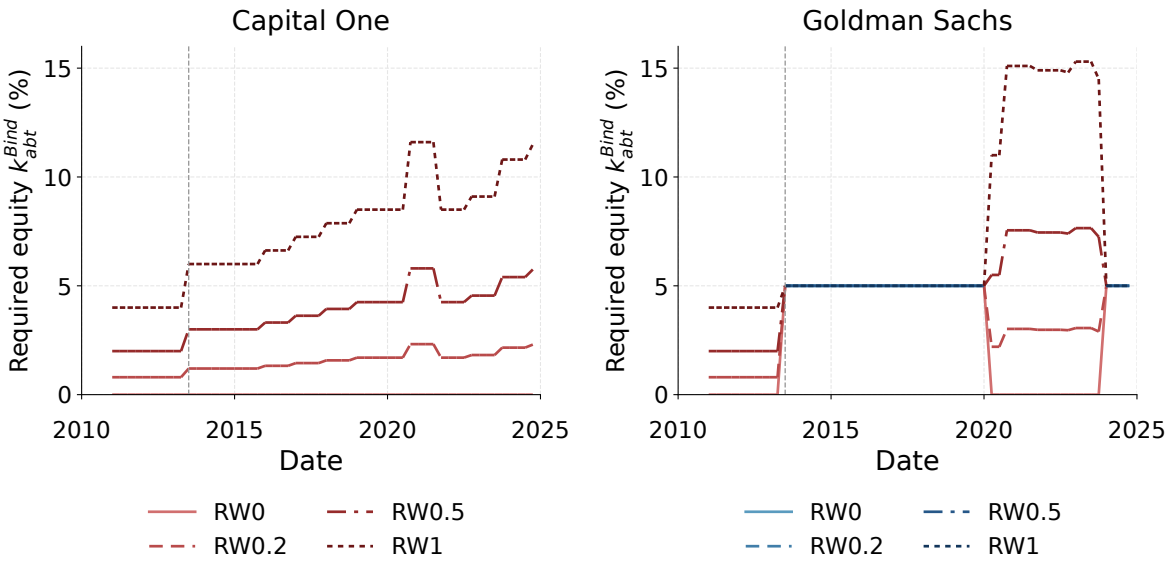


Figure 3: Required equity by asset over time: Capital One and Goldman Sachs. [Back to text]

Notes: Lines show the percentage of funding for assets in each risk-weight (RW) category that must come from equity (Tier 1 capital) under the bank's binding capital rule. Under a binding risk-based capital requirement, required equity equals the bank-level required ratio times RW. Under a binding supplementary leverage ratio (SLR), the required equity funding share for a nonexempt asset equals the bank's required SLR ratio. The bank's binding rule is the rule that requires the most Tier 1 capital given its portfolio composition and regulatory exposures. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

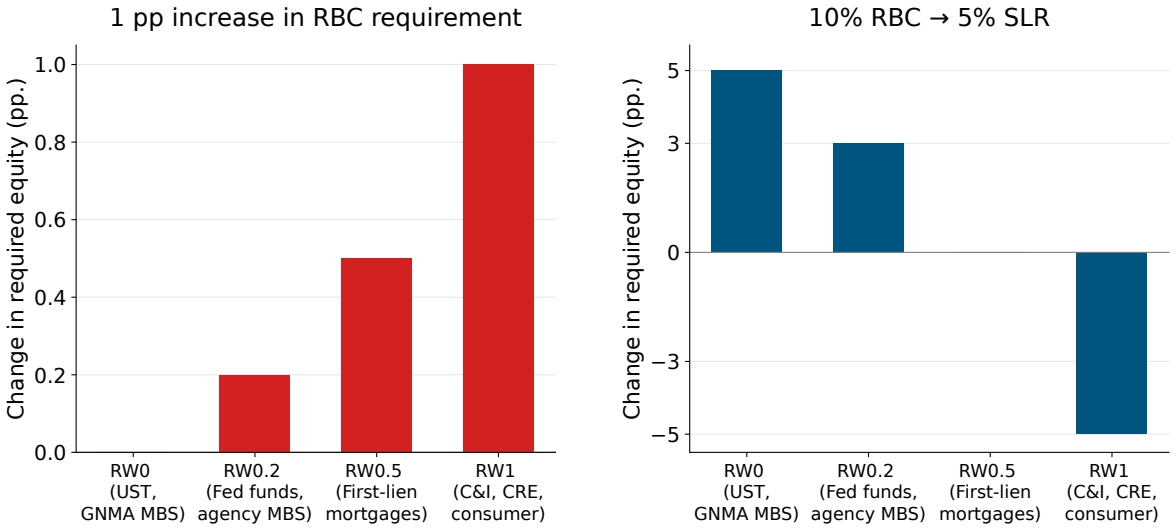


Figure 4: Changes in required equity across assets under an RBC increase and an RBC-to-SLR switch. [Back to text]

Notes: Bars show how required equity in each risk-weight (RW) category changes under two hypothetical regulatory changes, in percentage points. The left panel raises a binding risk-based capital (RBC) requirement by one percentage point, assuming RBC remains binding. The right panel replaces a binding 10% RBC requirement with a binding 5% supplementary leverage ratio (SLR) requirement.

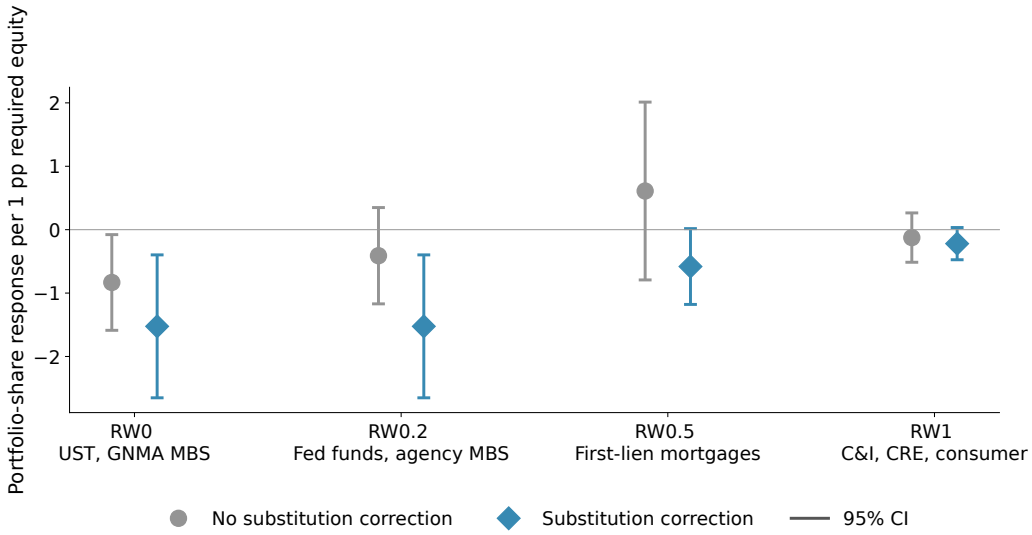


Figure 5: Long-run portfolio-share responses to higher required equity. [Back to text]

Notes: Points show portfolio-share responses (percentage points) to a one-percentage-point increase in required equity, pooling quarters 16–20. Gray circles omit substitution correction; blue diamonds include it and impose a common elasticity for RW0 and RW0.2. Error bars show 95% bank-clustered confidence intervals. Estimates use portfolio shares for 28 banks and include asset–date–horizon fixed effects. Portfolio shares exclude trading assets, reserves, and vault cash from the denominator; the uncorrected specification also includes bank–date fixed effects. RW denotes risk weight. Appendix Table C.8 reports the corresponding elasticities and standard errors. *Sources:* FR Y-9C bank regulatory filings.

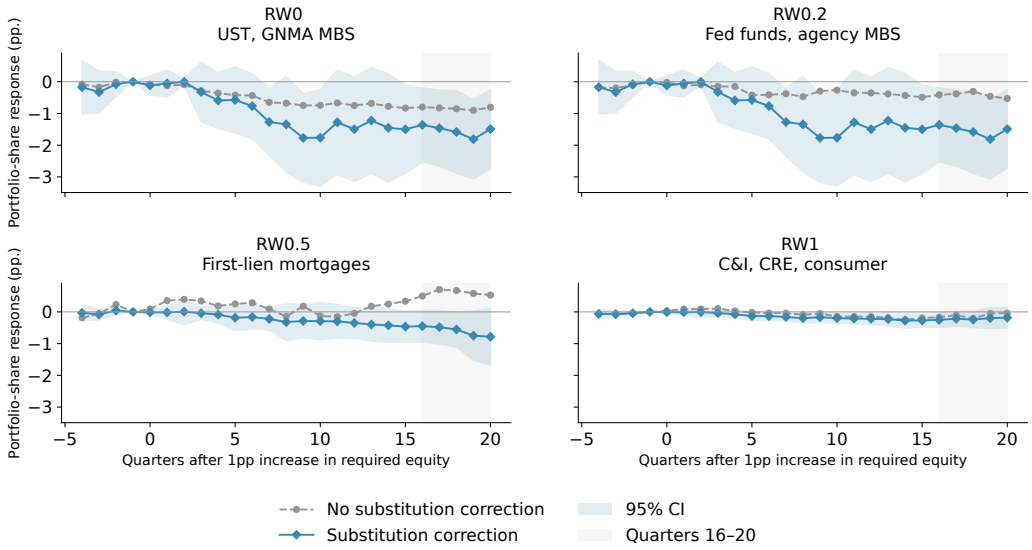


Figure 6: Dynamic asset-share responses with and without portfolio-substitution correction. [Back to text]

Notes: Lines show portfolio-share responses, in percentage points, to a one-percentage-point increase in required equity. Dashed gray lines with circles show estimates without substitution correction; solid blue lines with diamonds show estimates with portfolio-substitution correction that impose a common supply elasticity for RW0 and RW0.2 assets. RW denotes risk weight. Blue shading gives 95% confidence intervals clustered by bank. Quarter −1 is the zero reference point; quarter 0 includes the event-quarter response. Light-gray shading marks quarters 16–20, pooled in the long-run estimates. Estimates use portfolio shares for 28 banks, excluding trading assets, reserves, and vault cash from the denominator, and include asset–date–horizon fixed effects. The uncorrected specification also includes bank–date fixed effects. *Sources:* FR Y-9C bank regulatory filings.

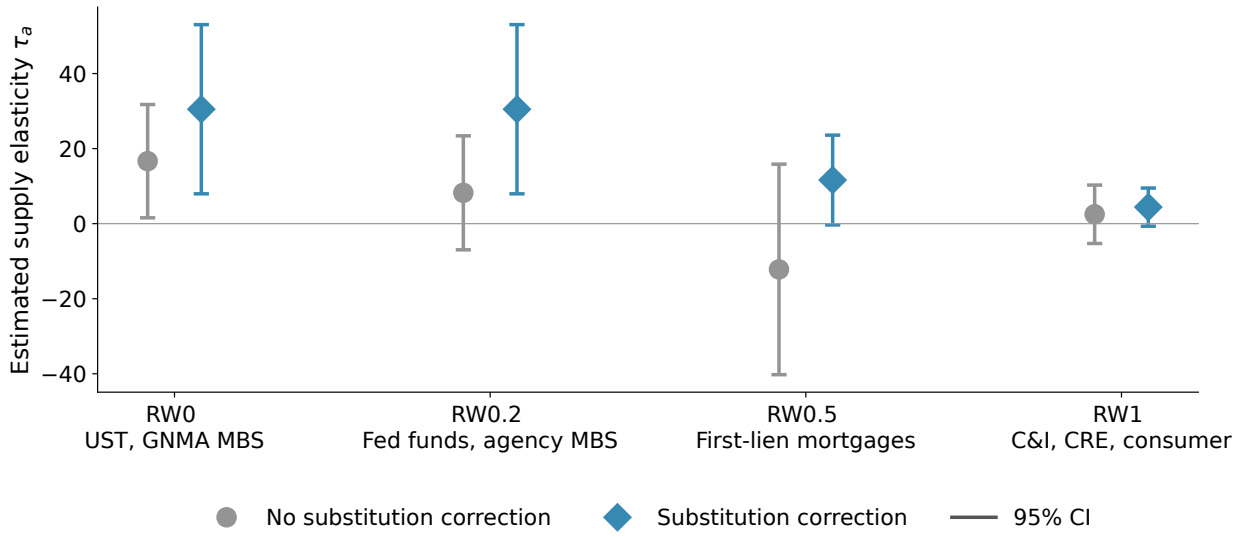


Figure 7: Long-run implied asset-supply elasticities. [Back to text]

Notes: Points show supply elasticities, $\widehat{\tau}_a = -\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the long-run portfolio-share response to required equity and $c_E = 0.05$ is the equity funding premium. Gray circles omit substitution correction; blue diamonds include the correction and impose a common supply elasticity for RW0 and RW0.2 assets. RW denotes risk weight. Error bars give 95% confidence intervals clustered by bank, treating c_E as fixed. Appendix Table C.8 reports the elasticities and standard errors. Estimates use portfolio shares for 28 banks, excluding trading assets, reserves, and vault cash from the denominator, and include asset–date–horizon fixed effects. The uncorrected specification also includes bank–date fixed effects. Long-run estimates pool quarters 16–20 after the change in required equity. *Sources:* FR Y-9C bank regulatory filings.

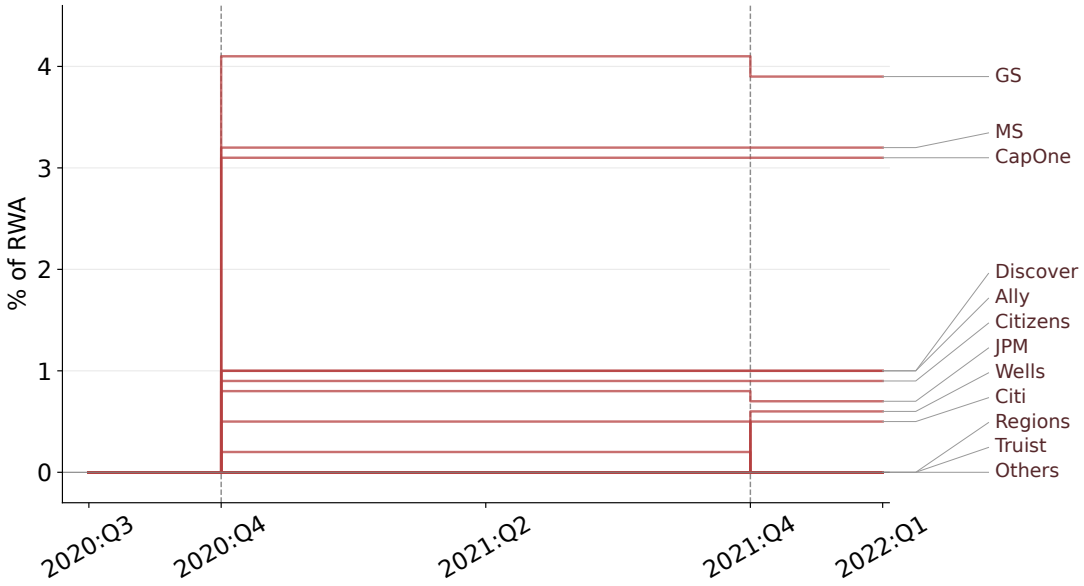


Figure 8: Stress Capital Buffer by bank in 2020 and 2021. [Back to text]

Notes: Lines cumulate each bank's Stress Capital Buffer changes in 2020 and 2021, in percentage points of risk-weighted assets (RWA). Bank abbreviations label the lines at the right; Others denotes banks with no buffer change. Dashed vertical lines mark 2020:Q4 and 2021:Q4, when the new buffers took effect. *Sources:* Federal Reserve capital-requirement announcements for 2020 and 2021.

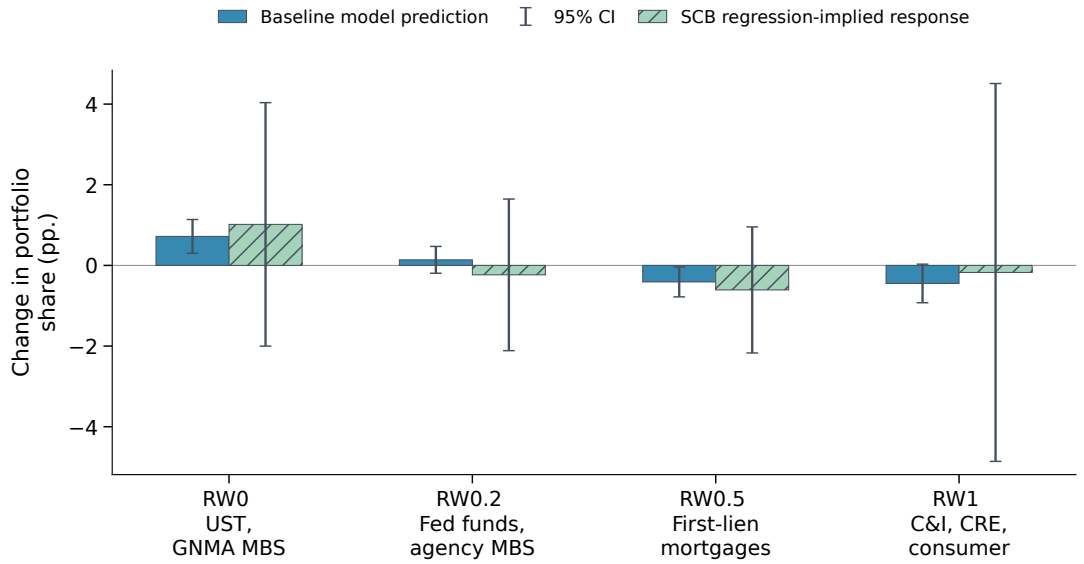


Figure 9: Predicted and estimated portfolio-share responses to SCB changes. [Back to text]

Notes: Blue bars (Baseline model prediction) apply the supply elasticities estimated in Section 4.1 to changes in required equity caused by Stress Capital Buffer (SCB) changes. Hatched bars (SCB regression-implied response) apply elasticities estimated from SCB changes to the same changes in assets' equity requirements. Bars show percentage-point changes in the portfolio share of each risk-weight (RW) category. The SCB regression pools available horizons between quarters 16 and 20. Responses are averaged over bank–event–horizon observations with nonzero SCB changes in 2020 and 2021; banks with no change remain in the regression comparison group. Both calculations account for portfolio substitution. Error bars show 95% delta-method confidence intervals using bank-clustered uncertainty in the corresponding supply-elasticity estimates. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve capital-requirement announcements for 2020 and 2021.

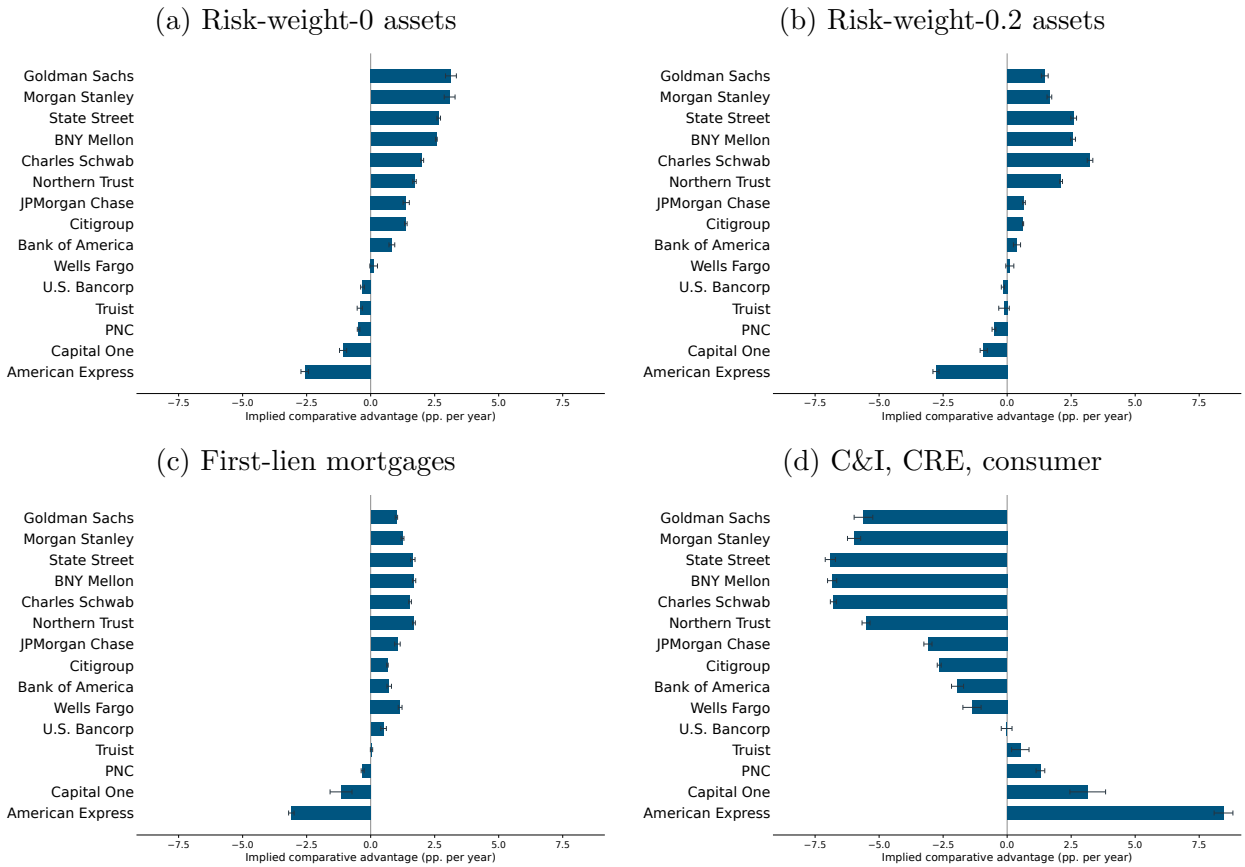


Figure 10: Implied comparative advantage across asset categories. [Back to text]

Notes: Bars show estimated bank-specific intercepts in marginal net returns, in percentage points per year, after accounting for equity requirements. The intercepts are normalized to average zero across banks within each asset and across assets within each bank. Error bars show 95% confidence intervals allowing for serial correlation across quarters, treating the equity funding premium and estimated supply elasticities as fixed. *Sources:* FR Y-9C bank regulatory filings.

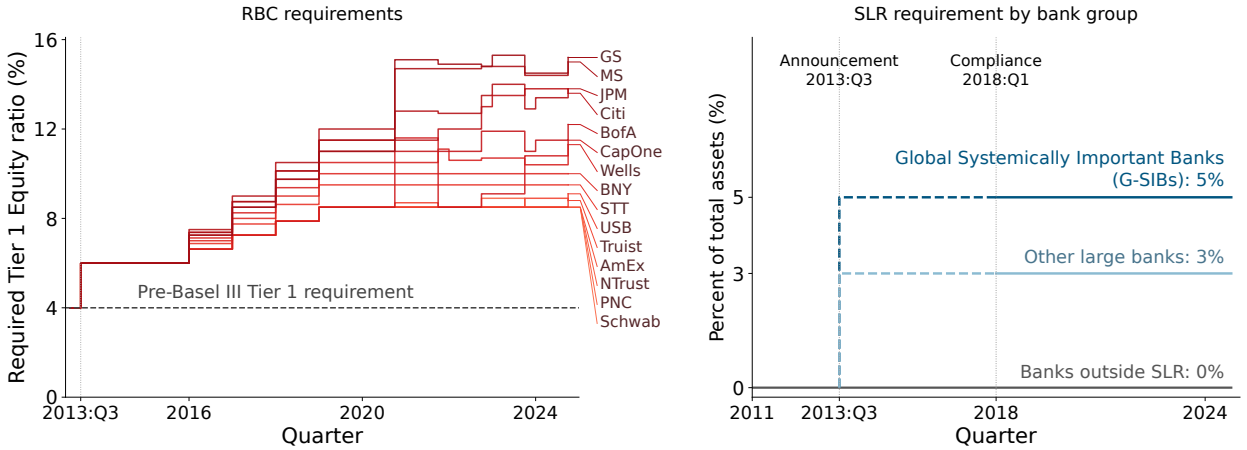


Figure 11: Historical bank-level RBC and SLR requirements. [Back to text]

Notes: The left panel shows each bank's required Tier 1 capital as a percentage of risk-weighted assets under risk-based capital (RBC) requirements. The right panel shows supplementary leverage ratio (SLR) requirements by bank group; G-SIB denotes a global systemically important bank. Vertical lines mark the SLR announcement and compliance dates. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

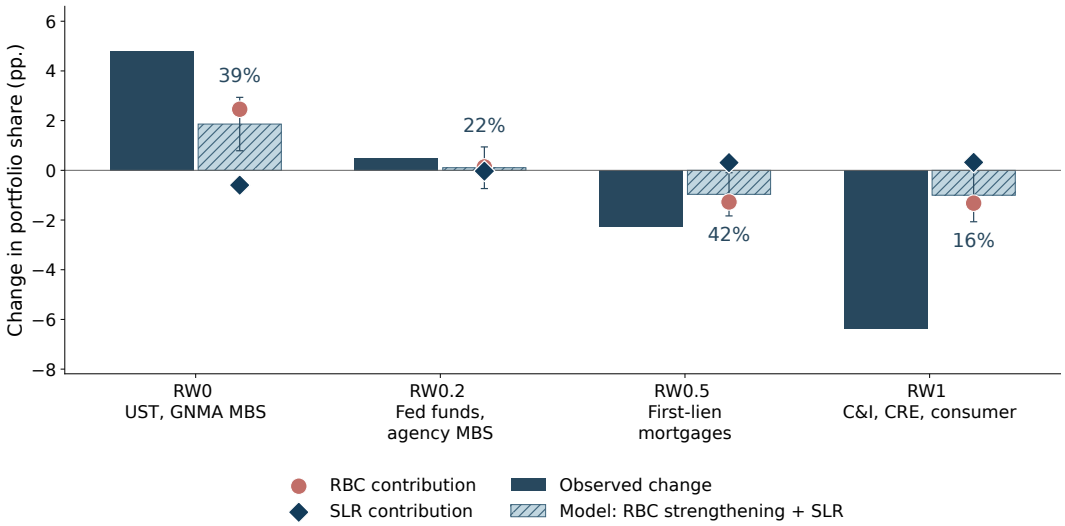


Figure 12: Observed aggregate portfolio shifts and predicted effects of RBC strengthening and SLR introduction. [Back to text]

Notes: Solid bars show observed portfolio-share changes from 2011:Q1–2013:Q2 to 2022–2024; hatched bars show predictions under actual requirements relative to pre-period RBC requirements without the SLR. Red circles show the RBC contribution; blue diamonds show the additional SLR contribution. Labels give the percentage of the observed change explained. Shares exclude trading assets, reserves, and vault cash. Bank weights and portfolio sizes are fixed at pre-period averages; observed and predicted changes use the same available quarters within each bank and comparison period. Error bars give 95% delta-method confidence intervals. RW denotes risk weight; RBC denotes risk-based capital; SLR denotes supplementary leverage ratio. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

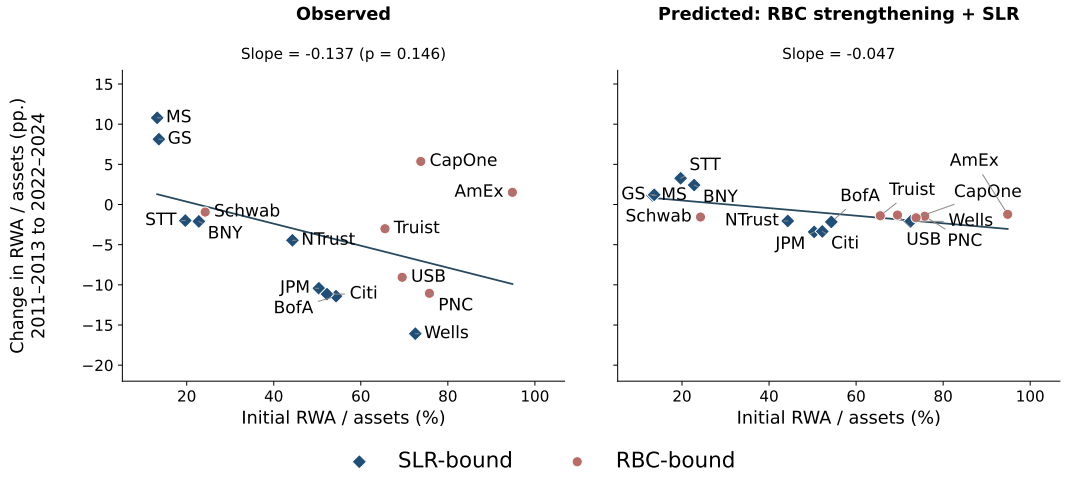


Figure 13: Observed and predicted convergence in banks' ratios of risk-weighted assets to assets. [Back to text]

Notes: Each point is a bank. RWA/assets is the average risk weight across the four asset categories, weighted by the dollar amount of assets in each category and excluding reserves and vault cash. Portfolio changes compare the 2011:Q1–2013:Q2 and 2022–2024 averages. Predictions hold each bank's total assets in the four categories fixed. Blue diamonds identify SLR-bound banks and red circles RBC-bound banks in 2013:Q4. Fitted lines weight banks equally; the observed-slope p-value uses heteroskedasticity-robust standard errors. RWA denotes risk-weighted assets; RBC denotes risk-based capital; SLR denotes supplementary leverage ratio. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

Appendix

A Model derivations and extensions

A.1 Proof of Proposition 1

Back

Indices a, b, and t denote assets, banks, and quarters. The portfolio share s_{abt} and equity ratio E_{bt} are measured relative to bank b's total assets. The symbols ℓ_{bt}^{RBC} and ℓ_{bt}^{SLR} denote required bank-level capital ratios, and w_a is asset a's risk weight. The equity funding premium is c_E, the supply elasticity is τ_a, and η_{bt} is the portfolio-constraint multiplier. The operator Δ denotes a change from quarter $t-1$ to quarter t.

For an interior portfolio, let λ_{bt}^{RBC} and λ_{bt}^{SLR} be the capital-constraint multipliers. The first-order condition for equity and complementary slackness for the RBC and SLR constraints give

$$\lambda_{bt}^{RBC} + \lambda_{bt}^{SLR} = c_E, \quad \lambda_{bt}^{RBC}\left(E_{bt} - \ell_{bt}^{RBC}\sum_a w_a s_{abt}\right) = 0, \quad \lambda_{bt}^{SLR}\left(E_{bt} - \ell_{bt}^{SLR}\sum_a s_{abt}\right) = 0.$$

When the RBC and SLR require different amounts of Tier 1 capital, the slack rule has multiplier zero and the binding rule has multiplier c_E. Differentiating the share choice in the constrained problem therefore gives the asset first-order condition in equation (3), with marginal required equity defined in equation (2).

The second-order approximation to asset a's net-return function r_{abt} around the fixed reference portfolio share $\bar{s}_{ab}$ is

$$r_{abt}(s) \approx r_{abt}(\bar{s}_{ab}) + r'_{abt}(\bar{s}_{ab})(s - \bar{s}_{ab}) - \frac{(s-\bar{s}_{ab})^2}{2\tau_a}.$$

Differentiating this displayed quadratic polynomial with respect to s gives

$$r'_{abt}(s) \approx r'_{abt}(\bar{s}_{ab}) - \frac{s - \bar{s}_{ab}}{\tau_a}.$$

Taking differences between dates $t-1$ and t, while holding the reference share $\bar{s}_{ab}$ fixed, gives $\Delta r\prime_{abt}(\bar{s}_{ab}) - \Delta s_{abt}/\tau_a$ for the change in marginal net returns along the chosen portfolios. Subtracting the asset first-order condition (3) at date $t-1$ from the condition at date t then yields

$$\Delta s_{abt} \approx \tau_a\big(\Delta r'_{abt}(\bar{s}_{ab}) - c_E \Delta k_{abt}^{Bind} - \Delta\eta_{bt}\big).$$

Holding net-return schedules fixed and summing across assets, the fixed portfolio total implies

$$0 = \sum_a \Delta s_{abt} \approx -c_E \sum_a \tau_a \Delta k_{abt}^{Bind} - \Delta\eta_{bt} \sum_a \tau_a, \qquad \Delta\eta_{bt} \approx -c_E \frac{\sum_a \tau_a \Delta k_{abt}^{Bind}}{\sum_a \tau_a}.$$

Substituting the expression for the change in the portfolio-constraint multiplier, $\Delta\eta_{bt}$, into equation (4), with net-return schedules held fixed, gives the portfolio-share response in equation (5). Approximation errors vanish to first order near the reference portfolio. Globally quadratic net returns make the marginal-net-return approximation exact.

A.2 From marginal net returns to the estimating equation

Back

Allowing marginal net returns to change, summing the asset-share first-order conditions and imposing the portfolio adding-up identity $\sum_a \Delta s_{abt} = 0$ gives

$$\Delta s_{abt} \approx \tau_a \left[\Delta r'_{abt}(\bar{s}_{ab}) - \frac{\sum_a \tau_a \Delta r'_{abt}(\bar{s}_{ab})}{\sum_a \tau_a} - c_E \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau} \right) \right]. \tag{7}$$

Assume that changes in marginal net returns at fixed reference shares satisfy

$$\Delta r'_{abt}(\bar{s}_{ab}) = \Delta \bar{r}_{at} + \Delta \bar{r}_{bt} + \Delta \varepsilon_{abt}.$$

The component $\Delta \bar{r}_{at}$ is common across banks holding asset a, $\Delta \bar{r}_{bt}$ is common across assets held by bank b, and $\Delta \varepsilon_{abt}$ is a bank–asset–date return shock. I assume that $\Delta \varepsilon_{abt}$ is uncorrelated with regulatory changes in equity requirements across the bank's assets, conditional on asset–date effects. Substituting this decomposition into equation (7) gives

$$\begin{aligned} \Delta s_{abt} \approx & \underbrace{\tau_a \left[\Delta \bar{r}_{at} - \frac{\sum_a \tau_a \Delta \bar{r}_{at}}{\sum_a \tau_a} \right]}_{\delta_{at}} + \underbrace{\tau_a \left[\Delta \bar{r}_{bt} - \frac{\sum_a \tau_a \Delta \bar{r}_{bt}}{\sum_a \tau_a} \right]}_{=0} \\ & - c_E \tau_a \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau} \right) + \underbrace{\tau_a \left[\Delta \varepsilon_{abt} - \frac{\sum_a \tau_a \Delta \varepsilon_{abt}}{\sum_a \tau_a} \right]}_{u_{abt}}. \end{aligned}$$

The underbraced term δ_{at} is the asset–date component of the portfolio-share response; u_{abt} is the remaining bank–asset–date component of the portfolio-share response. The bank–date return component $\Delta \bar{r}_{bt}$ is the same for every asset held by bank b. Its supply-elasticity-weighted average across assets, $\sum_a \tau_a \Delta \bar{r}_{bt} / \sum_a \tau_a$, therefore equals $\Delta \bar{r}_{bt}$ itself, so the bank–date component cancels from the portfolio-share response. With supply elasticities common

across banks, δ_{at} depends only on asset and date and is absorbed by asset–date fixed effects. After canceling the bank–date return component, the portfolio-share response becomes

$$\Delta s_{abt} \approx \delta_{at} - c_E \tau_a \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau} \right) + u_{abt}.$$

The model's portfolio-share response yields the local projection in equation (6).

A.3 Adjustment costs and long-run portfolio responses

Back

Portfolio adjustment costs can delay banks' responses to required equity without changing their long-run portfolio choices. Let $\mathbf{s}_{bt}$ be the vector of bank b's asset shares, and define the portfolio change as $\Delta \mathbf{s}_{bt} = \mathbf{s}_{bt} - \mathbf{s}_{b,t-1}$. The adjustment-cost function $G_b(\Delta \mathbf{s}_{bt})$ measures the cost of that reallocation per dollar of total assets. I assume that the adjustment-cost function is convex and continuously differentiable, with $G_b(\mathbf{0}) = 0$ and $\nabla G_b(\mathbf{0}) = \mathbf{0}$, where ∇G_b is the vector of marginal adjustment costs. Thus, leaving the portfolio unchanged incurs no adjustment cost, and the marginal adjustment cost is zero at an unchanged portfolio. The function can differ across banks and assets and depend jointly on changes in all asset shares; I do not impose a quadratic or other parametric form.

The bank maximizes the discounted sum of period profits from Section 2, subtracting the adjustment cost $G_b(\Delta \mathbf{s}_{bt})$ in each period, subject to the RBC and SLR capital constraints and a fixed portfolio total in each period. Let $0 < \rho < 1$ be its discount factor, and write $G_{b,a}$ for the partial derivative of the adjustment-cost function with respect to asset a's share change. At an interior optimum with one binding capital rule, the asset-share first-order condition is

$$r'_{abt}(s_{abt}) - c_E k_{abt}^{Bind} - \eta_{bt} = G_{b,a}(\Delta \mathbf{s}_{bt}) - \rho \mathbb{E}_t[G_{b,a}(\Delta \mathbf{s}_{b,t+1})]. \tag{8}$$

Here $\mathbb{E}_t$ denotes expectation conditional on information in quarter t. Increasing the current share incurs a marginal adjustment cost today and reduces the change needed to reach a given share next period. The right-hand side of equation (8) is the current marginal adjustment cost minus the discounted expected marginal adjustment cost saved next period.

Consider an unanticipated, permanent change in capital requirements from an initial stationary portfolio, holding net-return schedules and total portfolio size fixed. Suppose the optimal portfolio converges to an interior stationary portfolio and the expected marginal adjustment cost converges to zero. Continuity of ∇G_b and $\nabla G_b(\mathbf{0}) = \mathbf{0}$ imply that the right-hand side of equation (8) vanishes as adjustment ends. Once adjustment costs van-

ish, equation (8) reduces to the static asset first-order condition (3) used in Proposition 1. Linearizing marginal net returns around the initial portfolio gives the long-run response

$$\Delta s_{ab}^{LR} \approx -c_E \tau_a \left(\Delta k_{ab}^{Bind} - \frac{\sum_j \tau_j \Delta k_{jb}^{Bind}}{\sum_j \tau_j} \right).$$

Here Δs_{ab}^{LR} is the change in asset a's share between the initial and final stationary portfolios, and Δk_{ab}^{Bind} is the corresponding change in its binding equity requirement. Thus, a local projection at a horizon when adjustment is substantially complete recovers the static supply response without specifying the shape of adjustment costs. At shorter horizons, its coefficient also reflects the marginal adjustment costs in equation (8). The dynamic portfolio-share estimates in Figure 6 indicate whether the response has stabilized by quarters 16–20, the horizons pooled for long-run estimation.

B Data construction

B.1 Mergers

Back

Acquirer	Target	Close	Asset jump at close (%)
BB&T / Truist	SunTrust	2019:Q4	100
M&T	Hudson City	2015:Q4	26
M&T	People's United	2022:Q2	36
First Citizens	First Citizens Bcp	2014:Q4	37
First Citizens	CIT	2022:Q1	86
First Citizens	SVB	2023:Q1	96
BMO	M&I	2011:Q3	57
BMO	Bank of the West	2023:Q1	44
Huntington	FirstMerit	2016:Q3	36
Huntington	TCF Financial	2021:Q2	39
KeyCorp	First Niagara	2016:Q3	34
Charles Schwab	TD Ameritrade	2020:Q4	31
NYCB / Flagstar	Flagstar	2022:Q4	43
NYCB / Flagstar	Signature (deposits)	2023:Q1	37
Capital One	ING Direct USA	2012:Q1	43
Capital One	Discover	2025:Q2	34

Table B.1: Acquisitions triggering a bank-entity split [Back to text]

Notes: This table lists acquisitions for which banks are split into separate entities when the target exceeds 25% of the acquirer's assets at close. Asset jump at close is the acquirer's change in total assets from the preceding quarter to the closing quarter, as a percentage of preceding-quarter assets (FR Y-9C item BHCK2170). It measures the balance-sheet discontinuity removed by the split, not the target-to-acquirer size ratio. *Sources:* FR Y-9C bank regulatory filings (asset amounts).

B.2 Risk-weight mapping

Back

Asset	FR Y-9C mnemonic	w_a	12 CFR 217
Risk weight 0			
U.S. Treasury securities	BHCK0211, 1286	0	§32(a)(1)(i)
GNMA pass-through MBS	BHCKG300, G302	0	§32(a)(1)(i)
Risk weight 0.2			
Agency and GSE MBS and CMOs	BHCKG304, G306, G312, G314, G316, G318, K142, K144, K150, K152	0.2	§32(c)(1), §43
Agency and GSE non-MBS debt	BHCK1289, 1291, 1294, 1297, HT50, HT52	0.2	§32(c)(1)
Federal funds sold	BHDMB987	0.2	§32(d)(1)
Loans to U.S. depository institutions	BHCK1292	0.2	§32(d)(1)
Risk weight 0.5			
First-lien closed-end 1–4 family residential mortgages	BHDM5367	0.5	§32(g)(1)
Risk weight 1			
Commercial and industrial loans	BHCK1763, 1764, KX56	1	§32(f)(1)
Nonfarm nonresidential real-estate loans	BHCKF160, F161	1	§32(f)(1)
Junior-lien residential mortgages and home-equity lines	BHDM1797, 5368	1	§32(g)(2)
American Express charge-card reporting component	BHCKJ451 (Amex only)	1	§32(l)(5)
Credit-card loans	BHCKB538	1	§32(l)(5)
Other consumer loans	BHCKB539, K137, K207	1	§32(l)(5)
Agricultural loans	BHCK1590	1	§32(f)(1), (l)(5)
Finance lease receivables	BHDM2165	1	§32(l)(5)
Split across the classes by the bank's reported allocation			
Cash and depository balances (excluding the zero-weight portion)	BHCK0081, 0395	0.2, 0.5, 1	§32(d)
Securities purchased under agreements to resell	BHCKB989	0, 0.2, 1	§37

Table B.2: Asset reporting fields, standardized risk weights, and applicable regulatory provisions [Back to text]

Notes: FR Y-9C denotes the Consolidated Financial Statements for Holding Companies. Reporting-field identifiers specify the asset amounts used to construct each category; securities amounts are measured at amortized cost. The symbol w_a denotes asset a's standardized risk weight. GNMA denotes the Government National Mortgage Association; MBS denotes mortgage-backed securities; GSE denotes government-sponsored enterprise; CMO denotes collateralized mortgage obligation. *Sources:* FR Y-9C reporting instructions; 12 CFR Part 217.

Securities are measured at amortized cost, combining held-to-maturity and available-for-sale holdings. Legal references in Table B.2 refer to 12 CFR Part 217.

Mixed-weight categories. Agency non-MBS debt combines full-faith-and-credit obligations with GSE obligations, reported jointly from 2018:Q2; the category is assigned a

weight of 0.2 throughout. Federal funds sold are assigned 0.2. Private-label CMOs backed by agency collateral are assigned the 0.2 securitization floor (§217.43). First-lien residential mortgages are assigned 0.5, although Schedule HC-C does not identify all underwriting and performance conditions required for that weight.

Reverse repos. The 2015 revision to Schedule HC-R removed the risk-weight breakdown used to allocate reverse repos across asset categories. Before 2015, the reported breakdown combined reverse repos and federal funds sold. I use this breakdown to approximate the reverse-repo allocation. To construct a consistent series across the reporting change, I apply each bank's mean risk-weight fractions from 2011–2014 to its reverse-repo holdings in every quarter. For banks without usable historical reports, I use the sample's repo-weighted allocation. Holding these fractions fixed prevents the change in reporting from generating portfolio reallocation in the measured asset shares. The constructed series captures changes in total reverse-repo holdings while holding their risk-weight composition fixed. The regulatory treatment depends on collateral and counterparty characteristics (§217.37).

Cash and trading assets. I exclude Federal Reserve balances, vault cash, and all trading assets from both modeled asset holdings and the total assets used to calculate portfolio shares. Trading Treasuries have a zero specific-risk weight and are subject to the general market-risk capital requirement assessed on the trading portfolio (§§217.205, 217.210).

Loan coverage. Loans to nondepository financial institutions (BHCKJ454) and loans for purchasing or carrying securities (BHCK1545) enter the share denominator through total loans (BHCK2122) but are not assigned to a modeled risk-weight category. Other loans reported in BHCKJ451 also remain outside the modeled categories, with an exception for the American Express reporting correction described in Appendix B.

Cash and depository balances. Cash and balances at depository institutions can receive different risk weights. I use the fractions reported in Schedule HC-R for each bank and quarter to allocate its cash and U.S. depository balances (BHCK0081 and BHCK0395) across risk-weight categories. When a quarter's allocation is missing, I use the bank's mean fractions over quarters with reported allocations. If the bank has no usable reported allocation, I use the average fractions across reporting banks and quarters, weighted by cash and U.S. depository balances. I exclude the zero-weight portion from modeled holdings and from the total assets used to calculate portfolio shares. The portions assigned weights of 0.2, 0.5, and 1 enter the corresponding modeled categories. The portion assigned a weight of 1.5 remains in the total assets used to calculate portfolio shares but outside the modeled categories.

B.3 Data cleaning

Back

American Express. In 2016:Q3, American Express's charge-card receivables moved from HC-C item 9.b.(2), BHCKJ451 ("All other loans, excluding consumer loans"), into loan categories already included in risk weight 1. I include Amex's J451 holdings in risk weight 1 throughout the sample as a separate component to maintain consistent coverage across this reporting change. Including these holdings reduces the apparent discontinuity in C&I and other consumer loans from approximately $45.6 billion to $0.36 billion. Total loans already include these holdings, so the correction leaves the total assets used to calculate portfolio shares unchanged. The Federal Reserve's June 2017 FR Y-14 reporting notice documents updated guidance moving consumer charge cards from item 9.b.(2) to item 6.d; the corporate component of the reclassification is inferred from the reported balances. J451 holdings at other banks remain outside the modeled risk-weight categories.

Discover. Discover reported $10.7 billion of borrowed mortgage-backed securities in 2020:Q3 from a temporary securities-lending transaction that matured in November 2020. I subtract this amount from Discover's agency MBS holdings and from the total assets used to calculate its portfolio shares in 2020:Q3. The amount is the rounded carrying value and fair value of the collateral reported in Discover's 2020:Q3 Form 10-Q, Note 2, Investments.

Binding-rule smoothing. I smooth the quarterly binding-rule classifications using three adjustments in sequence. First, if a bank is classified under rule A, then rule B for two quarters, and then a different rule C, I assign those two quarters to rule C: A–B–B–C becomes A–C–C–C. Second, when a bank returns to its previous rule after at most four quarters under another rule, I assign the previous rule to those intervening quarters and repeat this step until no further changes occur. Spells lasting at least three quarters that begin in 2020:Q2–2021:Q1 are preserved. Third, I remove one- or two-quarter deviations between spells of the same rule that each last at least three quarters; the intervening quarters may be classified under different rules. These adjustments use both preceding and subsequent classifications.

B.4 Estimating banks' SLRs before reporting began

Back

Leverage exposure. Banks began reporting the SLR on the FR Y-9C in 2016:Q3. For 2013:Q1–2016:Q2, I construct a proxy by dividing reported Tier 1 capital by estimated total leverage exposure. Estimated exposure is the sum of on-balance-sheet assets used for the leverage ratio, credit-equivalent derivatives exposures, other off-balance-sheet credit-equivalent exposures, repo-style counterparty exposures, and an allowance for commitments

omitted from the risk-based exposure measure. Tables B.3 and B.4 list the reporting items and their construction. Missing component amounts are treated as zero when summed. The proxy is available for banks with reported SLRs in the calibration period.

Repo-style exposure. Before 2015, I impute repo-style counterparty exposure by multiplying gross repo-style balances by the bank's mean ratio of BHCKS516 to those balances over 2015:Q1–2016:Q2. Gross balances include federal funds sold and purchased, reverse repos, and repos. If the bank lacks a usable ratio, I use BHCE3433. From 2015:Q1 onward, I use BHCKS516 directly. In 224 bank-quarters over 2016:Q3–2019:Q4, the imputed and reported exposures have a correlation of 0.967; aggregate imputed exposure divided by aggregate reported exposure is 0.881.

Reporting changes. The derivatives exposure definition changes in 2015:Q1. I rescale the pre-2015 derivatives series by the ratio of mean derivative exposure relative to on-balance-sheet assets in 2015:Q1–Q2 to the corresponding mean in 2014:Q3–Q4. I also check new-format off-balance-sheet items for reporting breaks during 2015:Q2–2016:Q2. A break must exceed 5% of the preceding quarter's constructed leverage exposure both in the quarterly change and in the difference between the available four-quarter means before and after the break. This procedure identifies Morgan Stanley's BHCKG619 in 2016:Q1. For Morgan Stanley's earlier quarters under the revised reporting format, I replace BHCKG619 with its mean ratio to on-balance-sheet assets in 2016:Q1–Q2, multiplied by contemporaneous on-balance-sheet assets.

Calibration. I infer reported leverage exposure from Tier 1 capital divided by the reported SLR. For each bank, I average the ratio of reported to constructed exposure over 2016:Q3–2019:Q4 and multiply constructed exposure by this factor in every quarter. The proxy omits a separate adjustment for written credit derivatives because purchased protection needed to calculate the net exposure is unavailable in the extract. The bank-specific ratio of reported to constructed leverage exposure adjusts the proxy for average differences between the two exposure measures. I assess fit over the calibration period and in a held-out comparison that estimates the factor over 2016:Q3–2017:Q4 and evaluates it over 2018:Q1–2019:Q4.

FR Y-9C inputs. Table B.3 identifies the FR Y-9C inputs used to construct the SLR proxy. Dollar amounts are in the FR Y-9C reporting units; the proxy SLR is 100 times Tier 1 capital divided by calibrated leverage exposure.

Commitments. The commitment adjustment uses Schedule HC-L before 2015 and the revised Schedule HC-R from 2015 onward. The pre-2015 calculation treats credit-card commitments as unconditionally cancelable and applies a larger conversion factor to the remaining short-term commitments.

Component	Reporting items and treatment
Tier 1 capital	BHCK8274 through 2013:Q4; BHCA8274 from 2014:Q1.
On-balance-sheet assets	BHCKA224 through 2013:Q4; BHCAA224 from 2014:Q1.
Reported SLR	BHCAH036; positive reported values provide the calibration benchmark.
Derivatives	BHCEA167 through 2014:Q4; BHCKS542 + BHCKS549 from 2015:Q1. The earlier series is rescaled at the reporting change.
Other off-balance-sheet exposures, through 2014:Q4	Sum of BHCEB546, BHCE6570, BHCE3411, BHCE3429, BHCEA250, BHCEB541, BHCEB675, BHCEB681, BHCE6572, and BHCEG591.
Other off-balance-sheet exposures, from 2015:Q1	Sum of BHCKD992, BHCKD998, BHCKG607, BHCKG613, BHCKG619, BHCKS526, and BHCKG625. Morgan Stanley's BHCKG619 is adjusted as described above.
Repo-style gross balances	Sum of BHDMB987, BHCKB989, BHDMB993, and BHCKB995.
Repo-style counterparty exposure	BHCKS516 from 2015:Q1. Earlier exposure is imputed using the 2015:Q1–2016:Q2 mean exposure-to-gross-balances ratio; BHCE3433 is the fallback.

Table B.3: FR Y-9C inputs for Tier 1 capital and leverage exposure. [Back to text]

Notes: The supplementary leverage ratio (SLR) divides Tier 1 capital by total leverage exposure. The alphanumeric identifiers name the fields in the FR Y-9C bank regulatory filings used to measure each component. *Sources:* FR Y-9C bank regulatory filings.

Component	Reporting items and treatment
Total unused commitments, U	Sum of BHCK3814, BHCKJ455, BHCKJ456, BHCK3816, BHCK6550, BHCK3817, BHCKJ457, BHCKJ458, and BHCKJ459.
Short-term and cancelable commitments, S	$\max\{U - \text{BHCK6572} - \text{BHCKG591}, 0\}$. The subtracted items cover commitments over one year and commitments to ABCP conduits.
Cancelable portion, C	$\min\{\text{BHCKJ455} + \text{BHCKJ456}, S\}$.
Addition through 2014:Q4	$0.1C + 0.2(S - C)$.
Addition from 2015:Q1	$0.1 \times$ BHCKS540. Other reported credit-equivalent commitments enter the off-balance-sheet sum in Table B.3.

Table B.4: Commitment adjustment in the SLR exposure proxy. [Back to text]

Notes: Reported commitments from the FR Y-9C are converted to leverage-exposure amounts using the indicated conversion factors. The adjustment enters the denominator of the supplementary leverage ratio proxy. The symbols U, S, and C denote total unused, short-term and cancelable, and cancelable commitments, respectively. ABCP denotes asset-backed commercial paper. *Sources:* FR Y-9C bank regulatory filings.

Capital basis in earnings-call transcripts. I use disclosures from earnings-call transcripts identified as applying the final SLR rule. For each bank with overlapping disclosures,

I estimate the fully phased-in capital adjustment by dividing the difference between the reported FR Y-9C SLR and the earnings-call SLR by the fraction of deductions not yet phased in, then averaging across overlap quarters. The fractions of capital deductions not yet phased in are 0.8 in 2014, 0.6 in 2015, 0.4 in 2016, and 0.2 in 2017. I add the estimated adjustment times the remaining fraction to the earnings-call SLR. Disclosures already on a transitional basis enter unchanged; comparisons requiring an unavailable adjustment are omitted.

Goldman Sachs and Morgan Stanley. Before 2016:Q3, I multiply these banks' calibrated FR Y-9C proxies by the ratio of the adjusted earnings-call SLR to the proxy in quarters with usable earnings-call disclosures. I interpolate the factor linearly between quarters with earnings-call disclosures. Before the first quarter with an earnings-call disclosure, I use the mean factor from the first two quarters with earnings-call disclosures; after the final quarter with an earnings-call disclosure and before 2016:Q3, I retain the last factor. From 2016:Q3 onward, the binding-rule calculation uses the reported SLR.

Fit to reported SLRs. The calibrated proxy has a mean absolute error of 0.06 percentage points over 2016:Q3–2019:Q4. Calibrating only on 2016:Q3–2017:Q4 gives a mean absolute error of 0.07 percentage points in the held-out 2018:Q1–2019:Q4 observations.

C Calibration and additional estimation results

C.1 Equity funding cost calibration

Back

Table C.1 summarizes the literature used to calibrate the equity funding premium c_E. The assumed funding premium of $c_E = 0.05$ implies 5 basis points of additional funding cost per percentage point of required equity. Published loan-rate estimates range from 2.5 to 13 basis points and differ in assumptions about Modigliani–Miller offsets, taxes, and pass-through to loan rates. Loan-rate effects equal funding-cost effects under full pass-through; the estimates therefore provide calibration evidence rather than estimates of an identical parameter across studies.

Source	Loan-rate increase (basis points per pp of capital)	Assumptions
Kashyap et al. (2010)	2.5–4.5	Partial Modigliani–Miller offset; tax wedge
This paper ($c_E = 0.05$)	5	Funding cost before pass-through to loan rates
Miles et al. (2013)	≈ 5.5	45% Modigliani–Miller offset
Firestone et al. (2017)	3.9–7.8	50%–100% pass-through to loan rates
Brooke et al. (2015)	5–10	$\approx 53\%$ Modigliani–Miller offset
BCBS (2010)	13	No Modigliani–Miller offset

Table C.1: Published estimates used to calibrate the equity funding premium. [Back to text]

Notes: The assumed equity funding premium $c_E = 0.05$ is the additional annual cost of equity relative to debt and implies 5 basis points of additional funding cost per percentage point of required equity. Loan-rate effects equal funding-cost effects under full pass-through; estimates differ in assumptions about taxes, Modigliani–Miller offsets, and pass-through.

C.2 Tests of equal supply elasticities

Back

Null hypothesis	Baseline p-value
All three elasticities are equal	0.068
Securities = mortgages	0.118
Securities = RW1 loans	0.031
Mortgages = RW1 loans	0.206

Table C.2: Tests of equality of asset supply elasticities. [Back to text]

Notes: Tests use the baseline substitution-corrected specification, which pools RW0 and RW0.2 securities and estimates separate elasticities for RW0.5 mortgages and RW1 loans. RW denotes risk weight. Estimates pool quarters 16–20 and include asset–date–horizon fixed effects. The joint test imposes equality of all three elasticities; each pairwise test imposes equality of the two named elasticities. Tests use bank-clustered inference with 28 banks. Pairwise p-values are unadjusted for multiple testing. *Sources:* FR Y-9C bank regulatory filings.

C.3 Pre-trends in portfolio shares

Back

Figure C.1 extends the portfolio-share response estimates to quarters before the change in required equity. The pre-event coefficients test for portfolio-share changes preceding the regulatory event; a lack of significant pre-event coefficients does not establish the identifying assumption.

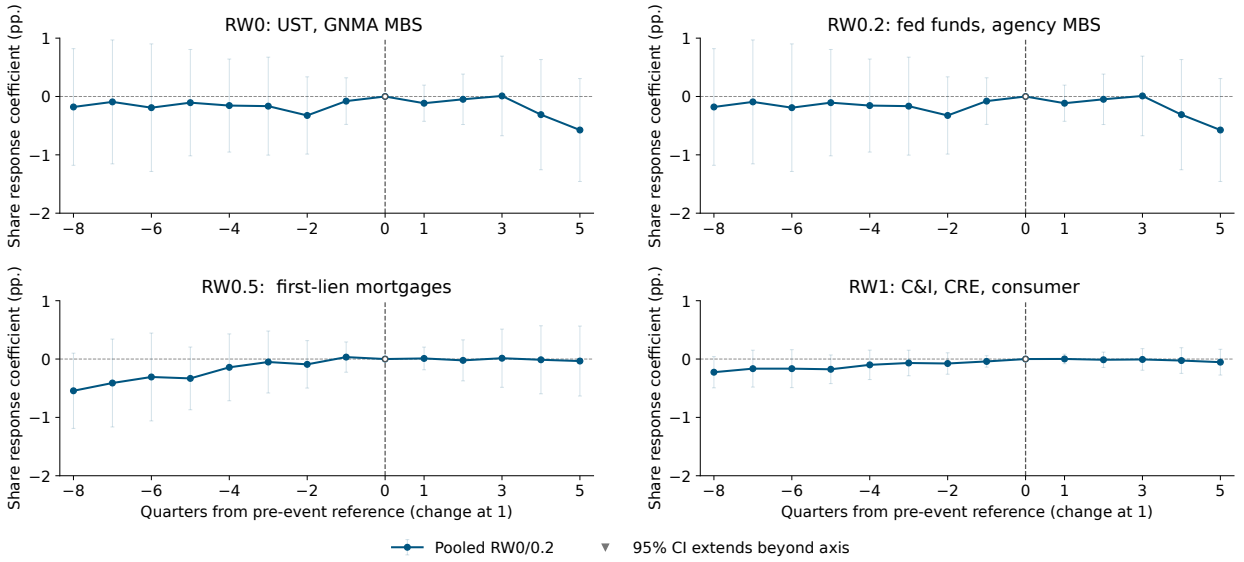


Figure C.1: Pre-period portfolio-share responses under discrete binding rules. [Back to text]

Notes: Points show portfolio-share response coefficients in each risk-weight category before and after a change in required equity. The substitution-corrected specification pools elasticities for risk weights 0 and 0.2. The regulatory event occurs in quarter t. The horizontal axis counts quarters from the reference quarter $t-1$, shown at 0 with a zero coefficient. The required-equity change occurs at 1; negative values precede the reference quarter. Coefficients measure percentage points of portfolio share per percentage-point increase in required equity. Error bars show 95% confidence intervals clustered by bank; triangles mark intervals extending beyond the axis. *Sources:* FR Y-9C bank regulatory filings.

C.4 Controls for lagged portfolio shares

Back

Table C.3 compares controls for the starting portfolio share, $s_{ab,t-1}$, with controls for four lagged shares, $s_{ab,t-1}, \ldots, s_{ab,t-4}$. After the full-sample baseline, the remaining columns use identical observations. Comparing these columns distinguishes conditioning on the bank's initial allocation from conditioning on its recent portfolio trajectory. The final column also controls for past required-equity changes in four-quarter bins.

	Baseline	Baseline common sample	Ctrl. for s_{t-1}	Ctrl. for $s_{t-1}, \ldots, s_{t-4}$	Ctrl. for $s_{t-1}, \ldots, s_{t-4}$ + 4q policy bins from $t-8$ to $t-1$
RW0 / 0.2	30.49***	27.17**	21.56*	22.22*	24.17**
	(10.99)	(11.13)	(10.91)	(10.90)	(10.72)
RW0.5	11.62*	10.66*	8.56	10.45*	11.22**
	(5.83)	(6.01)	(5.56)	(5.62)	(5.30)
RW1	4.41*	5.94**	5.24**	5.03*	5.88**
	(2.48)	(2.61)	(2.54)	(2.73)	(2.70)
Observations	17596	13108	13108	13108	13108
Banks	28	27	27	27	27
Adjusted R^2	0.13	0.17	0.24	0.26	0.26

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.3: Substitution-corrected supply elasticities with lagged portfolio-share controls. [Back to text]

Notes: RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. Columns 2–5 use a common sample. In the headings, $s_{t-1}, \ldots, s_{t-4}$ denote the asset's portfolio shares in the four quarters before regulatory-event quarter t; 4q denotes four-quarter intervals, and Ctrl. denotes controls. The last column adds required-equity changes over quarters 1–4 and 5–8 before the event. All control coefficients vary by asset. *Sources:* FR Y-9C bank regulatory filings.

C.5 Required equity evaluated on lagged portfolios

Back

Table C.4 evaluates old and new regulation on the same prior-quarter portfolio. Let $K_a(P, X)$ denote asset a's required equity when regulation is P and the binding rule is calculated using portfolio and regulatory exposures X. The constructed change is

$$Z_{abt} = K_a(P_t, X_{b,t-1}) - K_a(P_{t-1}, X_{b,t-1}).$$

Here Z_{abt} is the change in required equity for asset a at bank b in event quarter t, holding the preceding-quarter portfolio and regulatory exposures fixed. The calculation recomputes the binding rule under each regulation. A portfolio change alone generates no shock; a regulatory change can alter required equity with or without changing the binding rule. For each event quarter t, both terms in Z_{abt} use the portfolio and regulatory exposures observed at $t-1$. Changes in the portfolio between $t-2$ and $t-1$ do not enter Z_{abt} as a separate source of required-equity changes. Constructed shocks are not smoothed; actual required

equity retains the baseline smoothing. Policy dates, observed portfolio-share outcomes, and exclusions of SCB changes as starting events remain unchanged.

The final two columns of Table C.4 add past and then past-and-future policy controls to the lagged-portfolio specification. The controls for past required-equity changes use each earlier event's own prior-quarter portfolio. Future increments instead evaluate old and new regulation on the starting event's fixed portfolio $X_{b,t-1}$, so future portfolio responses do not enter these controls. Past required-equity controls sum changes over quarters 1–4 and 5–8 before the event. Future controls sum regulatory changes over successive four-quarter intervals after the event, ending at the response horizon. Each interval enters through a common component, a component proportional to the asset's risk weight, and an RW0-specific component, with coefficients varying by asset. All columns use their largest valid samples; differences across columns can reflect both specification and sample changes. These regressions replace actual required-equity changes with constructed regulatory changes; they are not instrumental-variable estimates.

	Baseline (actual required equity)	Required equity evaluated on lagged portfolio	Lagged portfolio Ctrl. for 4q bins $t-8$ to $t-1$	Lagged portfolio Ctrl. for 4q bins $t-8$ to $t+h$ Fixed future portfolio
RW0 / 0.2	30.49***	25.41**	20.84	18.42
	(10.99)	(11.07)	(12.46)	(15.19)
RW0.5	11.62*	8.66*	10.16	28.36
	(5.83)	(4.68)	(5.96)	(49.76)
RW1	4.41*	3.30	5.37*	8.61
	(2.48)	(2.34)	(2.78)	(5.60)
Observations	17596	17236	12748	12728
Banks	28	28	26	25
Adjusted R^2	0.13	0.14	0.19	0.22

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.4: Supply elasticities with actual and lagged-portfolio required equity, with and without policy controls. [Back to text]

Notes: In the headings, t is the regulatory-event quarter, h is the response horizon in quarters, 4q denotes four-quarter intervals, and Ctrl. denotes controls. RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. Columns 2–4 evaluate the event-quarter regulatory change on the preceding quarter's balance sheet, excluding switches caused solely by portfolio changes. Past controls use each earlier event's preceding-quarter balance sheet. Future controls use the starting event's preceding-quarter balance sheet. Required-equity changes are summed within four-quarter intervals. Each column uses its largest valid sample.
Sources: FR Y-9C bank regulatory filings.

C.6 Excluding pandemic SLR exemption start/end events

Table C.5 compares estimates including and excluding the introduction and expiration of the temporary SLR exemption, in 2020:Q2 and 2021:Q2. The exclusion removes these quarters as requirement-change dates; it retains other dates and their subsequent outcomes. The table reports the three-group specification. The substitution-corrected elasticities remain positive in every asset category and are larger when the exemption events are excluded.

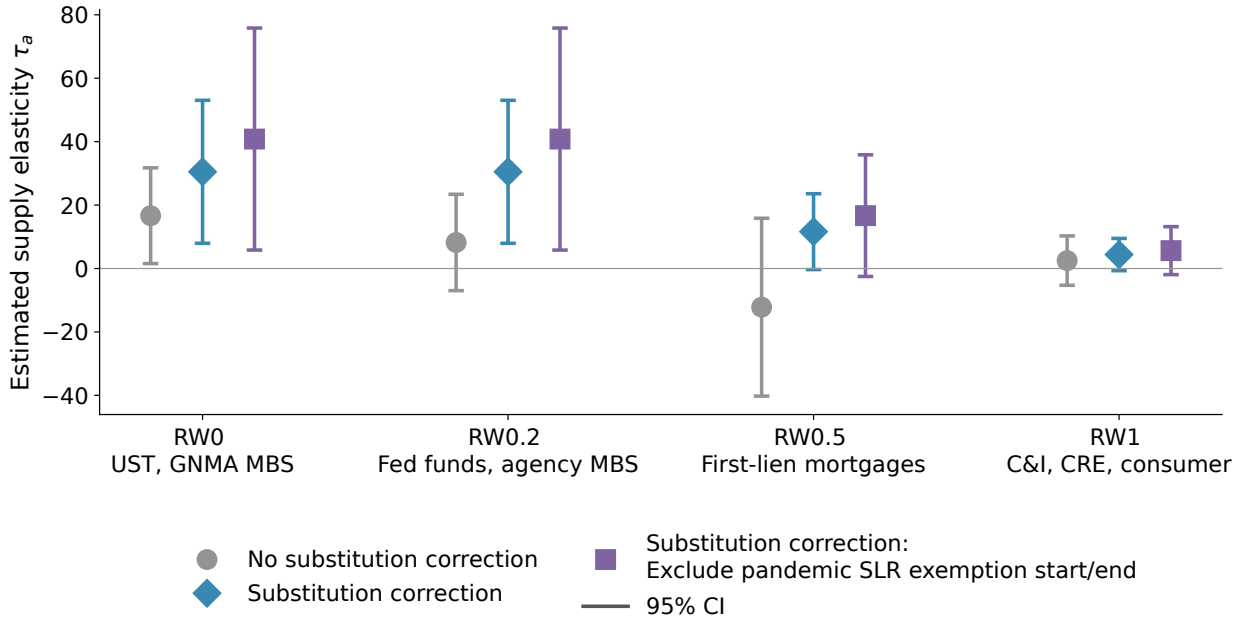


Figure C.2: Supply elasticities including and excluding pandemic SLR exemption events. [Back to text]

Notes: Gray circles and blue diamonds reproduce the main estimates without and with substitution correction. Purple squares apply the substitution correction after excluding 2020:Q2 and 2021:Q2 as requirement-change dates. Portfolio-share observations in those quarters remain available as outcomes for other starting dates. Corrected estimates pool RW0 and RW0.2; all estimates pool quarters 16–20 and use an equity funding premium of $c_E = 0.05$. Error bars are 95% bank-clustered confidence intervals. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

Asset	Include exemption events	Exclude exemption events
RW0	30.5***	40.8**
	(11.0)	(17.1)
RW0.2	30.5***	40.8**
	(11.0)	(17.1)
RW0.5	11.6*	16.7*
	(5.8)	(9.3)
RW1	4.4*	5.6
	(2.5)	(3.7)

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.5: Supply elasticities with and without pandemic SLR exemption events: discrete binding rules. [Back to text]

Notes: Estimates pool quarters 16–20. Include retains 2020:Q2 and 2021:Q2 as requirement-change dates; Exclude removes those two dates. Portfolio-share observations in those quarters remain available as outcomes for other starting dates. All specifications exclude reserves and vault cash. Both columns include asset–date–horizon fixed effects; the exclusion column also includes bank–date effects. The three-group specification pools RW0 and RW0.2 and estimates separate elasticities for RW0.5 and RW1. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

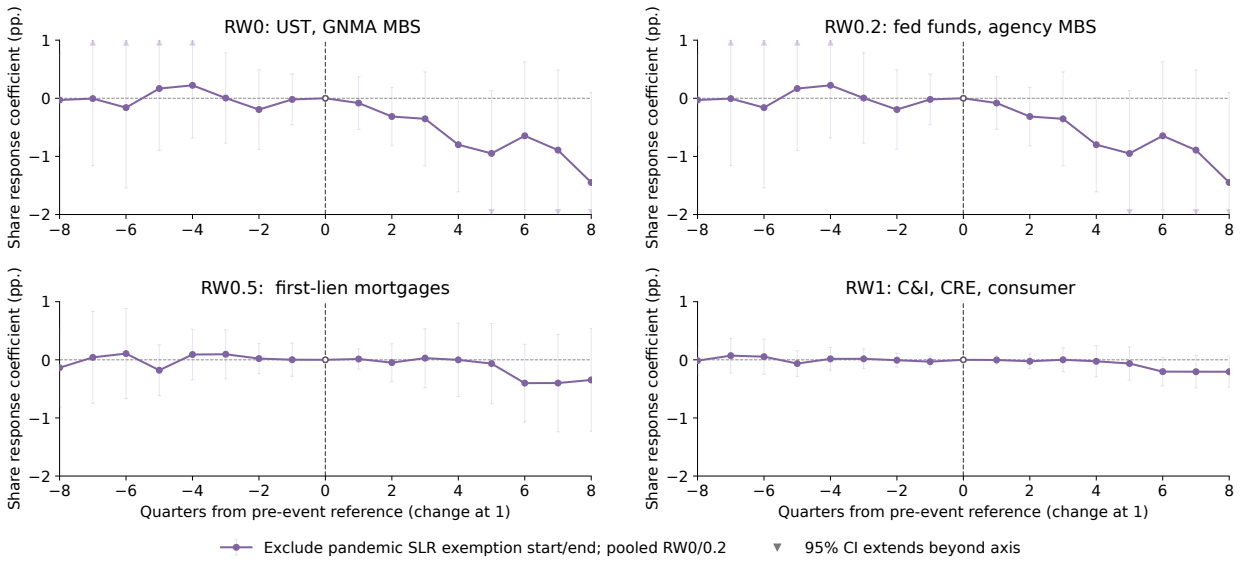


Figure C.3: Pre-trends excluding pandemic SLR exemption start/end events.

Notes: The specification excludes 2020:Q2 and 2021:Q2 as requirement-change dates, as in the exclusion column of Table C.5. Portfolio-share observations in those quarters remain available as outcomes for other starting dates. Substitution weights are proportional to the supply elasticities estimated after excluding those requirement-change dates. Quarter zero is the pre-event reference quarter; the required-equity change occurs at displayed quarter one. Error bars show 95% bank-clustered confidence intervals conditional on the substitution weights; triangles mark intervals extending beyond the axis. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

C.7 Controls for policy leads and lags

Back

Table C.6 adds controls for past required-equity changes in two bins, covering quarters 1–4 and 5–8 before the event. The final column also controls for future regulatory changes in successive four-quarter intervals through the response horizon, evaluating each change on the starting event's preceding-quarter portfolio. The event-quarter change and past controls continue to use actual required equity. Each four-quarter interval enters through a common component, a component proportional to the asset's risk weight, and an RW0-specific component, with coefficients varying by asset. The reported supply elasticities describe the response to the event-quarter change conditional on these policy controls.

	Baseline	Ctrl. for 4q bins from $t-8$ to $t-1$	Ctrl. for 4q bins from $t-8$ to $t+h$ Fixed future portfolio
RW0 / 0.2	30.49***	26.78**	24.07
	(10.99)	(11.44)	(14.79)
RW0.5	11.62*	12.21**	19.00
	(5.83)	(5.87)	(12.43)
RW1	4.41*	6.80**	8.74**
	(2.48)	(2.71)	(3.99)
Observations	17596	13168	12928
Banks	28	27	26
Adjusted R^2	0.13	0.18	0.22

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.6: Substitution-corrected supply elasticities controlling for leads and lags of required-equity changes. [Back to text]

Notes: In the headings, t is the regulatory-event quarter, h is the response horizon in quarters, 4q denotes four-quarter intervals, and Ctrl. denotes controls. RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. Event-quarter and past required-equity changes use contemporaneous portfolios. Future regulatory changes use the starting event's preceding-quarter balance sheet. Controls sum required-equity changes within four-quarter intervals. Each column uses its largest valid sample. *Sources:* FR Y-9C bank regulatory filings.

C.8 Alternative inference

Back

	Baseline	Bank WCB	Bank + quarter clustering	Bank + quarter WCB
RW0 / 0.2	30.49***	30.49*	30.49***	30.49*
SE	(10.99)	(10.99)	(8.48)	(8.48)
RW0.5	11.62*	11.62*	11.62**	11.62*
SE	(5.83)	(5.83)	(4.88)	(4.88)
RW1	4.41*	4.41	4.41***	4.41*
SE	(2.48)	(2.48)	(1.59)	(1.59)
Observations	17596	17596	17596	17596
Banks	28	28	28	28
Adjusted R^2	0.13	0.13	0.13	0.13

Standard errors clustered by bank or by bank and event quarter, as indicated, in parentheses. Bootstrap columns use bootstrap p-values.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.7: Alternative inference for the baseline three-group specification. [Back to text]

Notes: RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. All columns report the baseline estimates. WCB denotes restricted wild cluster bootstrap-t inference with 999 bank-level Rademacher draws and nonlinear refitting. Two-way covariance uses inclusion–exclusion, with negative eigenvalues set to zero; two-way WCB uses bank-and-starting-quarter studentization. *Sources:* FR Y-9C bank regulatory filings.

C.9 Alternative asset-pooling specifications

Back

The substitution-corrected estimates in Table C.8 use three alternative restrictions on the asset-specific response coefficients β_a. Since $\tau_a = -\beta_a/c_E$, each restriction also applies to the supply elasticities.

Pooling securities only. The specification used in the main results assigns a common coefficient to RW0 and RW0.2 assets and estimates separate coefficients for RW0.5 and RW1 assets:

$$\beta_{\text{RW0}} = \beta_{\text{RW0.2}} = \beta_L, \qquad \beta_{\text{RW0.5}} = \beta_M, \qquad \beta_{\text{RW1}} = \beta_H.$$

The coefficients β_L, β_M, and β_H correspond to securities, first-lien mortgages, and RW1 assets, respectively.

Pooling securities and pooling loans. The two-coefficient specification retains the common coefficient for RW0 and RW0.2 assets and also assigns a common coefficient to

RW0.5 and RW1 assets:

$$\beta_{\text{RW0}} = \beta_{\text{RW0.2}} = \beta_L, \qquad \beta_{\text{RW0.5}} = \beta_{\text{RW1}} = \beta_H.$$

Linear in risk weight. The linear specification estimates an intercept and a slope relating the response coefficient to the asset's risk weight w_a:

$$\beta_a = \beta_0 + \beta_1 w_a.$$

Here β_0 is the intercept, β_1 is the slope, and w_a is asset a's standardized risk weight. I report the converged solution with the lowest residual sum of squares among solutions with a positive sum of supply elasticities. Table C.8 compares supply elasticities under these three restrictions with estimates that omit the portfolio-substitution correction and allow a separate coefficient for each risk-weight category. Response coefficients satisfy $\widehat{\beta}_a = -c_E \widehat{\tau}_a$; their standard errors equal c_E times the reported elasticity standard errors.

	No substitution correction	Pooled RW0/0.2	Pooled RW0/0.2 and RW0.5/1	Linear $\beta_0 + \beta_1 RW$
RW0	16.65**	30.49***	31.36***	25.49***
	(7.36)	(10.99)	(11.25)	(8.38)
RW0.2	8.21	30.49***	31.36***	21.29***
	(7.40)	(10.99)	(11.25)	(6.76)
RW0.5	−12.19	11.62*	5.54**	14.99***
	(13.67)	(5.83)	(2.39)	(4.47)
RW1	2.50	4.41*	5.54**	4.49*
	(3.79)	(2.48)	(2.39)	(2.63)
N. Banks	28	28	28	28
N. Quarters	42	42	42	42
SE clustering	Bank	Bank	Bank	Bank
Adjusted R^2	0.13	0.11	0.11	0.11
Adjusted within R^2	0.00	0.00	0.00	0.00
Bank–date FE	Yes	No	No	No
Asset–date–horizon FE	Yes	Yes	Yes	Yes

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.8: Supply elasticities under alternative asset-pooling specifications. [Back to text]

Notes: RW denotes risk weight. Pooled RW0/0.2 imposes a common securities elasticity; two groups additionally pools RW0.5 and RW1. Linear in RW restricts elasticity to an intercept and slope in risk weight. No correction omits portfolio substitution. Estimates measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event. Elasticities equal $-\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the portfolio-share response coefficient and $c_E = 0.05$ is the equity funding premium. *Sources:* FR Y-9C bank regulatory filings.

C.10 Substitution into assets outside the four categories

Back

The four asset categories do not exhaust the banking-book denominator. The baseline substitution correction treats the remaining assets as having a zero supply response. Table C.9 checks whether the estimated elasticities depend on this restriction by adding those remaining assets as a fifth category, the outside good, and estimating its supply elasticity. All columns retain the baseline dollar amounts of assets in the four risk-weight categories and the same banking-book denominator, including its exclusion of reserves and vault cash. The four categories and the outside good therefore exhaust that denominator.

The first column of Table C.9 reports the baseline elasticities with a zero supply elasticity for the outside good. The two outside-good specifications measure a proxy for the outside good's average risk weight from FR Y-9C Schedule HC-R and allow its portfolio share to respond to regulatory costs. Both outside-good specifications calculate the outside good's

average risk weight as its risk-weighted assets divided by the dollar amount of assets in the outside good. They measure that average risk weight before the regulatory-change quarter and hold it fixed when calculating the change in required equity under the bank's binding rule. The substitution correction then includes the estimated outside elasticity. Treating the heterogeneous outside assets as one category remains an aggregation assumption: their average risk weight does not necessarily capture differences in supply responses across those assets.

Outside risk weight from asset components. I divide assets outside the four risk-weight categories into loans, securities, and cash balances. For omitted loans, I use the ratio of reported risk-weighted assets to carrying values in HC-R's two ordinary "all other loans" rows as a proxy for their average risk weight. For omitted securities, I subtract the estimated risk-weighted assets of the included securities from reported securities risk-weighted assets, including separately reported securitizations when available. The remaining cash balances use HC-R cash risk-weight allocations. Before 2015, I apply each bank's median component risk weights over 2015:Q4–2017:Q4 to its earlier component holdings, requiring at least four valid calibration quarters. This permits earlier composition changes but assumes that the component risk weights can be carried backward across the reporting change. From 2015 onward, I use the available lagged measurements directly.

Outside risk weight from reported RWA residual. I instead measure outside-loan risk-weighted assets by subtracting the estimated risk-weighted assets of the included loans from total reconstructed loan risk-weighted assets. Securities and cash follow the same approach as in the asset-component specification. This specification uses the available HC-R measurements in each reporting regime rather than carrying later component weights backward. It relies more directly on the accuracy of the included-loan mapping and on the coverage of earlier HC-R reports. Observations with negative loan or securities residuals are excluded. Both checks retain approximations in the securities mapping and, where necessary, the allocation of jointly reported securitization risk-weighted assets.

The estimates for the four asset categories remain close to the baseline re-estimated on the same observations in Panel B. The outside elasticities are positive but imprecisely estimated. The samples differ because the checks require usable outside-asset measurements. Relative to the baseline, the asset-component specification excludes SVB Financial Group, KeyCorp, and Huntington; the reported-RWA-residual specification excludes SVB Financial Group and KeyCorp. The securities risk-weighted-asset residuals exclude SVB Financial Group and KeyCorp. Huntington lacks enough valid 2015:Q4–2017:Q4 observations to impute its earlier component risk weights and has no usable later observations in the asset-component specification.

	Baseline: fixed outside good	Outside RW from asset components	Outside RW from reported RWA residual
A. Supply elasticities			
RW0 and RW0.2	30.49***	28.06**	32.28***
	(10.99)	(11.11)	(10.93)
RW0.5	11.62*	12.37*	11.91*
	(5.83)	(6.60)	(5.84)
RW1	4.41*	5.50**	4.43**
	(2.48)	(2.27)	(2.09)
Outside good	0 (fixed)	0.47	5.12
		(7.03)	(11.19)
Bank–quarter–horizon observations	4,399	3,656	3,468
Banks	28	25	26
B. Baseline on each column's sample			
RW0 and RW0.2	30.49***	27.99**	32.63***
	(10.99)	(11.05)	(10.56)
RW0.5	11.62*	12.51*	12.38*
	(5.83)	(6.85)	(6.05)
RW1	4.41*	5.54**	4.68*
	(2.48)	(2.59)	(2.31)

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.9: Supply elasticities allowing substitution into the outside good. [Back to text]

Notes: RW denotes risk weight; RWA denotes risk-weighted assets. The outside good comprises banking-book assets outside the four modeled risk-weight categories. All columns use the banking-book denominator, pool horizons 16–20, impose a common elasticity for RW0 and RW0.2, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects. Panel A fixes the outside elasticity at zero in the baseline and estimates it in the two outside-good specifications. Panel B re-estimates the original baseline on each column's sample. Observations count bank–quarter–horizon combinations, before stacking asset categories. Inference treats the measured risk-weight proxies and equity funding premium as fixed. *Sources:* FR Y-9C bank regulatory filings.

C.11 Risk-weight allocations reported in Schedule HC-R

Back

Table C.10 compares the baseline mapping with banks' reported risk-weight allocations in Schedule HC-R, Part II. I apply the reported fractions within ordinary loan and securities categories to the same modeled dollar amounts used in the baseline. Both specifications use identical observations, with regulatory-change quarters from 2015 onward and responses pooled over quarters 16–20. Reverse-repo allocations, the treatment of cash, the portfolio-share denominator, and asset-specific required equity retain their baseline construction. The comparison isolates differences in the risk-weight mapping within the modeled loan and securities categories.

	Category mapping	HC-R fractions
RW0 / 0.2	33.23***	37.23***
	(11.06)	(11.07)
RW0.5	8.36**	6.87**
	(3.29)	(2.53)
RW1	4.03*	4.85**
	(2.10)	(1.76)
Observations	10632	10632
Banks	28	28
Adjusted R^2	0.19	0.19

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.10: Supply elasticities using Schedule HC-R risk-weight allocations on a matched sample from 2015 onward. [Back to text]

Notes: Both columns use the three-group substitution correction and measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event. HC-R denotes the FR Y-9C regulatory-capital schedule; RW denotes risk weight. HC-R fractions allocate the baseline loan and securities amounts across risk weights. Ordinary HC-R securities exclude separately reported securitization exposures. Both columns pool RW0 and RW0.2, use the same post-2015 sample and quarters 16–20, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects. *Sources:* FR Y-9C bank regulatory filings.

C.12 Sensitivity to the equity funding premium

Back

Figure C.4 reports implied supply elasticities over alternative values of c_E. A larger equity funding premium implies a smaller supply elasticity for a given estimated portfolio-share response, since $\widehat{\tau}_a = -\widehat{\beta}_a / c_E$.

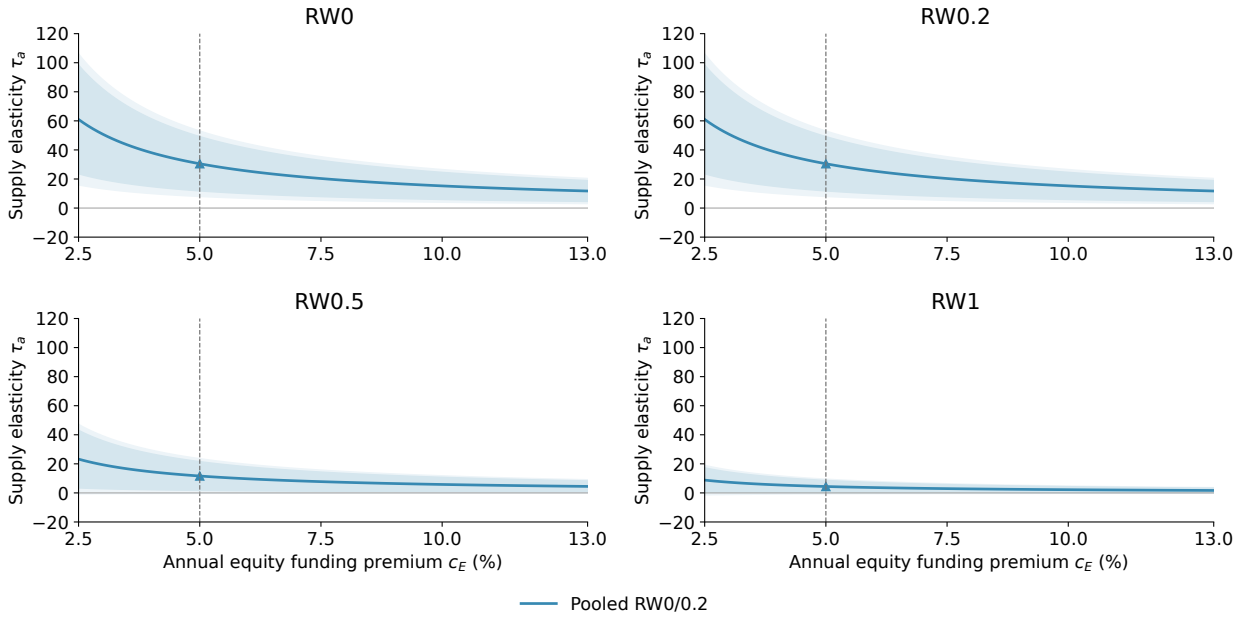


Figure C.4: Long-run supply elasticities under alternative equity funding premia. [Back to text]

Notes: Supply elasticities $\widehat{\tau}_a$ equal $-\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the long-run portfolio-share response coefficient and c_E is the equity funding premium. The horizontal axis expresses the equity funding premium c_E as a percentage; the baseline value is 5%. RW denotes risk weight. The three-group curves impose a common supply elasticity for RW0 and RW0.2 and separate elasticities for RW0.5 and RW1. Shaded bands show 90% and 95% confidence intervals clustered by bank, conditional on c_E. *Sources:* FR Y-9C bank regulatory filings.

C.13 Continuous weights on RBC and SLR requirements

Back

Banks may put weight on both capital rules when they are uncertain which will bind. I replace the required-equity measure based on a single binding capital rule with a weighted average of the RBC and SLR requirements:

$$k_{abt}^{Bind} = p_{bt}\ell_{bt}^{RBC}w_a + (1-p_{bt})\ell_{bt}^{SLR}, \qquad p_{bt} = \Phi(\Delta_{bt}/\sigma_\Delta).$$

Here k_{abt}^{Bind} is asset a's required equity at bank b in quarter t, w_a is the asset's risk weight, and ℓ_{bt}^{RBC} and ℓ_{bt}^{SLR} are the bank-level required ratios. The weight p_{bt} applies to the RBC requirement. The buffer difference Δ_{bt} is the SLR buffer minus the RBC buffer, with both buffers measured as excess equity per dollar of risk-weighted assets. The function Φ is the standard normal cumulative distribution function. I estimate the buffer-uncertainty parameter σ_Δ jointly with the supply elasticities. A larger buffer under SLR relative to RBC puts more weight on RBC; a larger σ_Δ makes the weights less sensitive to the difference between buffers.

The continuous-weight estimates preserve the main ranking: securities respond more to required equity than loans. The three-group elasticity is 47.1 for RW0 and RW0.2, 16.9 for RW0.5, and 4.9 for RW1, compared with 30.5, 11.6, and 4.4 under discrete binding rules. The continuous-weight estimates are less precise. Figures C.5 and C.6 compare response coefficients and supply elasticities with the discrete-rule baseline; Table C.11 reports the continuous-weight estimates and standard errors. Figure C.7 reports portfolio-share responses over time under continuous rule weights.

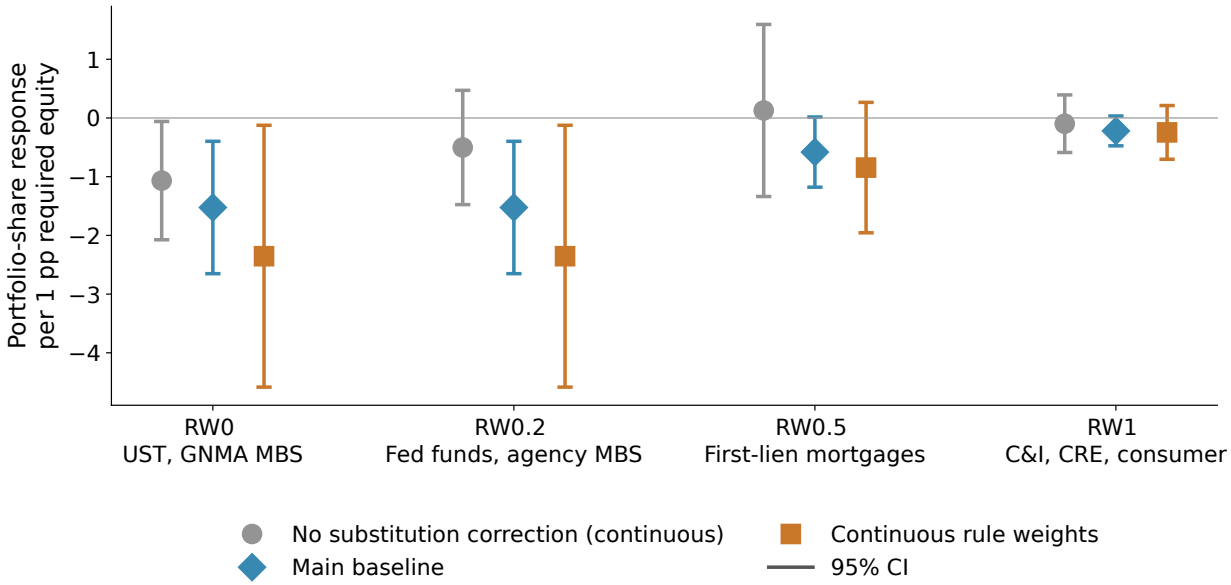


Figure C.5: Long-run portfolio-share responses with and without substitution correction under continuous rule weights. [Back to text]

Notes: Points show portfolio-share responses, in percentage points, to a one-percentage-point increase in required equity, pooling quarters 16–20. Blue diamonds show the main discrete-rule baseline. Orange squares show continuous-rule estimates with substitution correction and pooled RW0 and RW0.2; gray circles show continuous-rule estimates without substitution correction. Error bars give 95% bank-clustered confidence intervals. Both continuous regressions use the substitution-corrected estimate of the sensitivity of rule weights to RBC–SLR capital-buffer differences. Confidence intervals for the uncorrected regression treat this sensitivity as fixed. RBC denotes risk-based capital; SLR denotes supplementary leverage ratio; RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

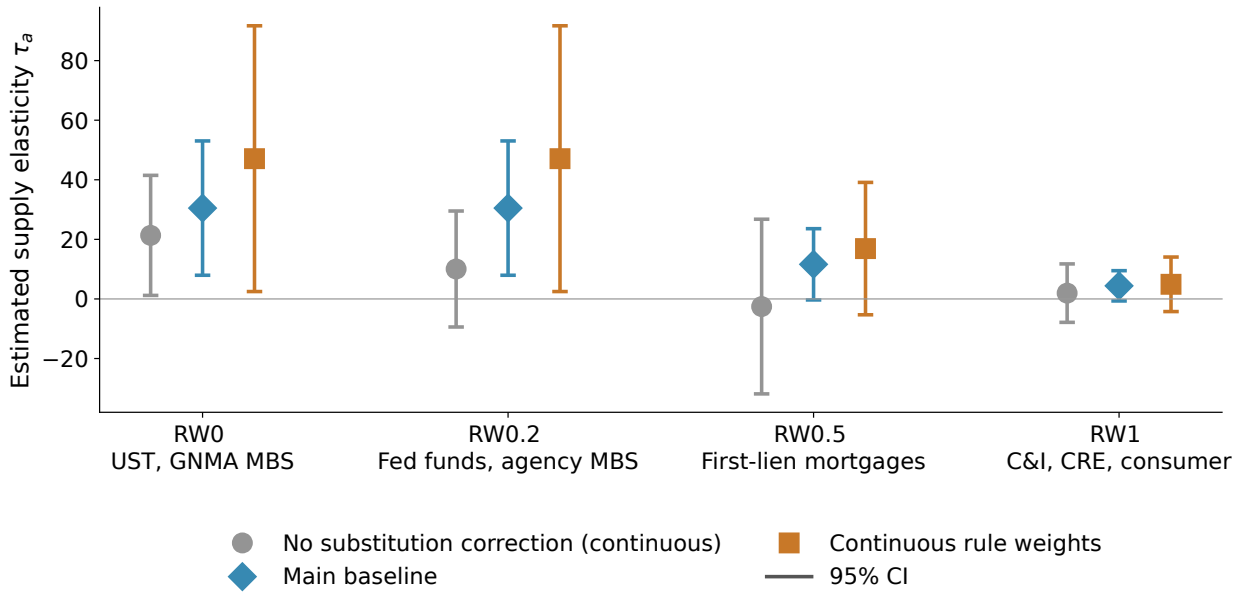


Figure C.6: Supply elasticities with and without substitution correction under continuous rule weights. [Back to text]

Notes: Points show supply elasticities $\widehat{\tau}_a$, pooling quarters 16–20 with an equity funding premium of 5%. Blue diamonds show the main discrete-rule baseline. Orange squares show continuous-rule estimates with substitution correction and pooled RW0 and RW0.2; gray circles show continuous-rule estimates without substitution correction. Error bars give 95% bank-clustered confidence intervals. Both continuous regressions use the substitution-corrected estimate of the sensitivity of rule weights to RBC–SLR capital-buffer differences. Confidence intervals for the uncorrected regression treat this sensitivity as fixed. RBC denotes risk-based capital; SLR denotes supplementary leverage ratio; RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

	Response coefficient $\widehat{\beta}_a$	Supply elasticity $\widehat{\tau}_a$
RW0 / 0.2	-2.35^{**}	47.09^{**}
	(1.09)	(21.74)
RW0.5	-0.84	16.90
	(0.54)	(10.83)
RW1	-0.25	4.91
	(0.22)	(4.46)
Observations	17596	17596
Banks	28	28
Adjusted R^2	0.13	0.13

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.11: Response coefficients and supply elasticities with continuous rule weights. [Back to text]

Notes: Estimates pool quarters 16–20 and impose a common elasticity for RW0 and RW0.2. Required equity weights the risk-based capital and supplementary leverage ratio requirements using the difference between their capital buffers. Supply elasticities $\widehat{\tau}_a$ equal $-\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the portfolio-share response coefficient and $c_E = 0.05$ is the equity funding premium. Inference conditions on c_E. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

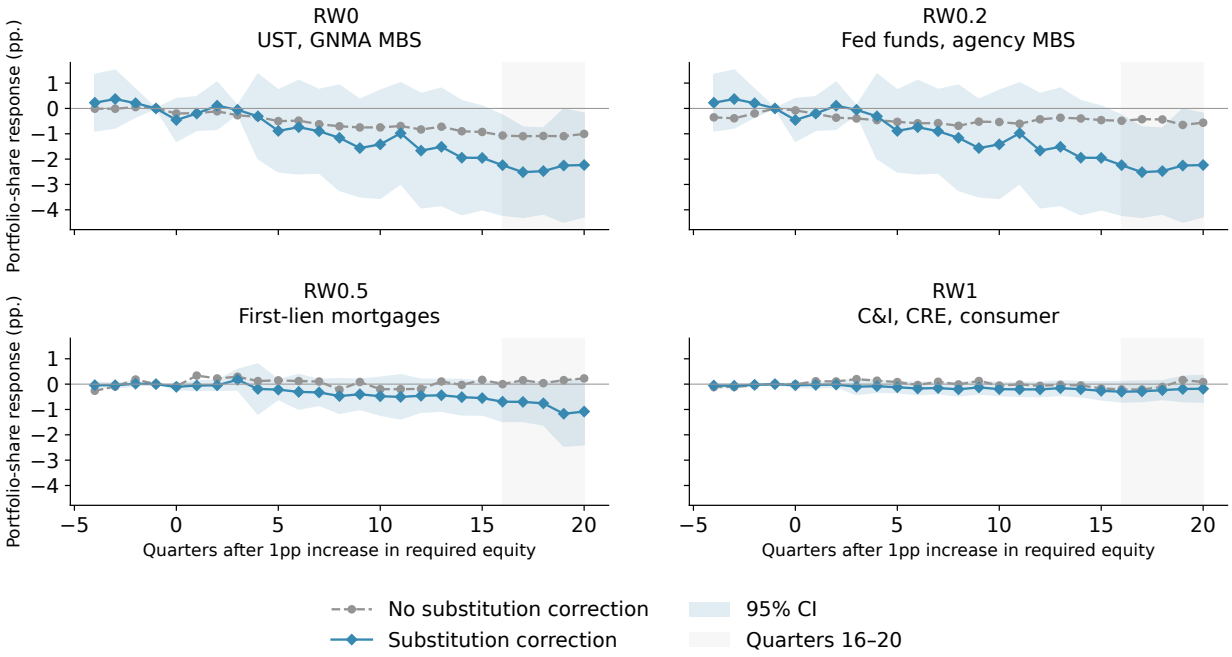


Figure C.7: Dynamic asset-share responses with continuous rule weights. [Back to text]

Notes: Lines show portfolio-share responses, in percentage points, to a one-percentage-point increase in required equity. Dashed gray lines with circles show estimates without substitution correction; solid blue lines with diamonds show estimates with portfolio-substitution correction that impose a common supply elasticity for RW0 and RW0.2 assets. RW denotes risk weight. Blue shading gives 95% confidence intervals clustered by bank. Quarter −1 is the zero reference point; quarter 0 includes the event-quarter response. Light-gray shading marks quarters 16–20, pooled in the long-run estimates. Required equity weights the risk-based capital (RBC) and supplementary leverage ratio (SLR) requirements using the difference between the bank's capital buffers under the two rules. Estimates use portfolio shares for 28 banks, excluding trading assets, reserves, and vault cash from the denominator, and control for bank–date and asset–date–horizon fixed effects. *Sources:* FR Y-9C bank regulatory filings.

D Uncertainty in the policy application

Table D.1 reports the observed and model-implied convergence slopes from Section 5. Each slope relates the change in a bank's average asset risk weight to its initial average risk weight, giving each bank equal weight. The percentage explained is 100 times the model-implied slope divided by the observed slope. Confidence intervals are reported for the underlying slopes; the percentage explained is reported as a point estimate.

For the aggregate portfolio changes and model-implied convergence slope, I apply the delta method to the full bank-clustered covariance matrix of the baseline supply elasticities. The 95% intervals use Student's t critical values with 27 degrees of freedom, holding historical portfolios, regulatory requirements, bank weights, the equity funding premium, and the application sample fixed. The observed-convergence interval uses heteroskedasticity-robust standard errors with the HC1 finite-sample correction and a Student's t critical value with 13 degrees of freedom.

	Observed convergence slope	Model-implied convergence slope	Percentage of convergence explained
Point estimate	-0.14	-0.05	34.6%
95% confidence interval	[-0.33, 0.05]	[-0.08, -0.01]	—

Table D.1: Observed and model-implied portfolio convergence. [Back to text]

Notes: The sample contains 15 banks with initial and final portfolio observations. Portfolio changes compare the 2011:Q1–2013:Q2 and 2022–2024 averages. Both regressions include an intercept and give each bank equal weight. Brackets report 95% confidence intervals. Observed-slope intervals use heteroskedasticity-robust standard errors; model-implied intervals reflect uncertainty in the bank-clustered supply-elasticity estimates. The percentage explained divides the model-implied slope by the observed slope. *Sources:* FR Y-9C bank regulatory filings.